

Do Retail Investors Drive Lottery-Like Stock Mispricing?

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Abstract

This paper studies whether retail participation explains the underperformance of lottery-like stocks. Prior research links retail trading to attention-driven buying after salient events. In contrast, we examine systematic retail participation. Using weekly Robinhood holdings data, we construct measures capturing persistent ownership and flow dynamics rather than short-lived attention shocks. We study their relation to the MAX anomaly, the negative relation between extreme returns and future returns. Systematic retail participation predicts positive future returns, while the MAX anomaly remains negative and economically significant. Retail participation does not strengthen the MAX effect, indicating distinct sources of return predictability.

Keywords: Systematic retail participation; Lottery-like stocks; MAX anomaly; Asset pricing; Return predictability; Robinhood holdings

1.Introduction

Financial markets frequently display pricing patterns that are difficult to reconcile with standard asset-pricing models. One of the most persistent and widely discussed examples is investors' attraction to lottery-like stocks that offer a small probability of extremely large returns. A rich body of literature documents that such stocks, despite being highly sought after by individual investors, generally underperform subsequently (Kumar, 2009; Bali, Cakici, and Whitelaw, 2011; Barberis and Huang, 2008). This pattern has led to the intuitive belief that retail investor demand contributes to the negative future returns of lottery-like securities, particularly because retail investors are often more prone to attention biases, sentiment-driven trading, and speculative tendencies compared to institutional investors. However, despite the intuitive rationale of this explanation, the empirical link between retail activity and the cross-sectional underperformance of lottery-like stocks remains theoretically asserted but not conclusively tested using investor-level data.

The growth of Robinhood Markets has created an opportunity to revisit this question. Robinhood provides a uniquely transparent view of retail investor participation, especially among younger, more speculative, and attention-driven traders. Multiple studies show that Robinhood users disproportionately trade in high-attention stocks, extreme-return stocks, and speculative securities (Barber et al., 2022). Prior research suggests that retail investors respond strongly to salient events and attention-grabbing signals. Evidence shows that large one-day returns stimulate retail buying (Seasholes and Wu, 2007; Greenwood and Nagel, 2009), that individual investors rely heavily on attention due to limited search capacity (Barber and Odean, 2008), and that attention-driven trading on platforms such as Robinhood often precedes negative future returns (Barber et al., 2022). Overall, the literature implies a causal narrative in which increased retail participation fuels speculative buying, contributing to overpricing and lower subsequent returns, particularly among lottery-like stocks.

Yet, despite the conceptual coherence of this argument, markets are more nuanced. Retail participation is multi-dimensional. Some flows represent impulsive, attention-driven speculation, while others reflect gradual accumulation, increased platform adoption, diversification, or non-speculative trading. As a result, not all retail activity should be expected to generate overpricing or negative future returns. Importantly, the retail proxies used in prior research, such as abnormal volume, social media attention, or inclusion on Robinhood's "Top Mover" lists, primarily capture attention shocks, not systematic participation. It therefore remains an open empirical question whether retail participation, measured outside of extreme, attention-driven events, systematically predicts the underperformance of lottery-like stocks.

This paper revisits the relationship between lottery-like stock performance and retail investor activity using weekly Robinhood holdings data that allow us to measure retail participation more precisely and at a higher frequency than prior studies. Unlike the literature that relies low-frequency monthly snapshots or attention-based proxies, we construct four distinct weekly retail

activity measures, each capturing a different economic dimension of retail behavior: (i) net retail buying pressure, based on week-to-week changes in the number of investors holding each stock (adjusted for shares outstanding); (ii) abnormal retail ownership, which identifies stocks held more or less heavily by retail traders relative to firm size and price; (iii) abnormal retail flows, which isolate unexpected shifts in investor holdings after controlling for size and price; and (iv) a microstructure-adjusted retail flow measure, which further removes variation associated with short-term volatility, liquidity, and recent return dynamics. Because these measures are constructed from holdings rather than trades and vary at the weekly frequency, they capture persistent retail involvement rather than isolated trading episodes.

Throughout the paper, we refer to this form of retail activity as systematic retail participation. By this, we mean slow-moving, stock-level retail ownership and accumulation patterns that persist across weeks and are not mechanically tied to transitory attention shocks, short-term liquidity demand, or extreme return realizations. This concept stands in contrast to the attention-driven retail trading emphasized in much of the existing literature, which focuses on sharp, short-lived inflows following salient events such as news, viral price moves, or platform-driven visibility. Distinguishing systematic participation from attention shocks is crucial, as prior research predominantly links negative future returns to the latter. An open empirical question, therefore, is whether persistent retail participation, as opposed to episodic speculative bursts, exhibits different return implications and whether it interacts with established lottery-like anomalies such as the MAX effect.

We begin by confirming that the MAX anomaly, the negative relationship between the maximum daily return in the prior period and future returns (Bali et al., 2011), holds clearly in weekly data. Using standard quintile sorts, we observe that high-MAX stocks underperform low-MAX stocks both in equal-weighted and value-weighted portfolios, and across holding periods ranging from one to eleven weeks. These results validate that our setting captures the traditional lottery-like stock pricing pattern.

Next, we examine whether retail participation measures explain or interact with the MAX anomaly. Following the intuition of the existing literature, one might expect retail activity to intensify the negative MAX effect. If retail investors drive the overpricing of lottery-like stocks, stocks with both high MAX exposure and high retail participation should yield the lowest future returns. Surprisingly, this is not what we find.

Across all four retail activity measures, retail participation robustly predicts positive future returns, not negative. Retail-heavy stocks outperform retail-light stocks over the subsequent week. This result is consistent across all measures, all weeks, and both equal- and value-weighted portfolios. Rather than indicating speculative mispricing, our retail signals capture persistent demand pressure that is not attributable to attention shocks, liquidity, volatility, or short-term return dynamics.

The key finding emerges when we conduct double-sorted portfolios on retail activity and MAX. We first sort stocks into quintiles based on a retail activity measure, and then, within each retail quintile, further sort stocks into MAX quintiles. Across all portfolio constructions, the MAX effect

remains negative within every level of retail participation. Moreover, the spread between high-MAX and low-MAX stocks is statistically similar across retail activity quintiles. These results indicate that the intensity of retail participation neither strengthens nor weakens the MAX anomaly. In particular, even stocks with heavy retail involvement do not exhibit stronger underperformance of high-MAX names relative to those with low retail presence. These results show that MAX underperformance and retail participation operate independently in the weekly setting. The MAX anomaly does not become more severe when retail investors are more active, nor does greater retail presence offset or reverse the effect. Retail traders, therefore, do not appear to be the marginal price-setters responsible for the observed MAX underperformance. In other words, lottery pricing and retail activity are not mechanically linked in the manner widely assumed in the existing literature.

To ensure that our results are not driven by earnings-related return jumps, a central concern raised by Nguyen and Truong (2018), we split the sample into earnings-announcement (EA) weeks and non-EA (NOEA) weeks. The results remain qualitatively identical. In EA weeks, where price reactions reflect information rather than speculation, retail participation is even more positively related to returns, while the MAX anomaly continues to hold. In NOEA weeks, where speculation might be more influential, the same independence between retail activity and MAX persists.

The evidence challenges the prevailing assumption that retail demand is the source of the MAX anomaly or the underperformance of lottery-like stocks. Instead, our results suggest weekly retail participation, when measured through holdings data rather than attention proxies, predicts positive returns and does not interact meaningfully with MAX. This distinction reconciles the literature. Retail attention shocks, captured by extreme news, viral events, or abnormal volume, may indeed cause overpricing and negative returns. But systematic weekly participation, measured through holdings in a granular platform like Robinhood, does not.

Our findings contribute to the literature in several ways. First, we provide the multi-dimensional weekly decomposition of retail activity using four structural measures derived directly from investor holdings. Second, we show that retail participation is not the mechanism underlying the MAX anomaly, contradicting the dominant narrative but consistent with the recognition that not all retail activity is speculative. Third, we document a robust positive relation between retail activity and future returns, suggesting that retail investors in this setting may act as informed or at least liquidity-providing participants rather than uninformed noise traders. Finally, our results highlight that different forms of retail activity must be carefully distinguished and that attention-driven retail demand behaves very differently from persistent retail participation.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature and outlines the main hypotheses. Section 3 describes the data and variable construction, and Section 4 includes methodology and definitions of the four retail measures. Section 5 presents the main empirical results for MAX, retail activity, and their interaction. Section 6 concludes.

2. Related Literature and Motivation

The relationship between retail investor participation and the return patterns of lottery-like stocks lies at the intersection of two substantial literatures, retail investor behavior and the pricing of speculative, high-skew assets. Prior work implicitly assumes that retail investors contribute to the mispricing of lottery-like securities, yet empirical tests rarely rely on direct holdings data and rarely use stock-level retail ownership measures. The following sections review these two literatures and outline the conceptual gap that motivates the current analysis.

2.1. Retail Investor Behavior: Attention vs. Systematic Participation

The dominant narrative in the literature characterizes retail investors as attention-driven, reactive, and prone to speculative trading. Barber and Odean (2008) provide foundational evidence that individual investors disproportionately buy stocks that stand out on salient dimensions such as extreme one-day returns, abnormal volume, or news coverage. Their argument centers on the presence of search frictions. Retail investors face limitations in their ability to screen the full universe of stocks, leading their trading activity to concentrate on securities that temporarily attract increased attention. In contrast, institutional investors, who face fewer attention constraints and employ more systematic stock selection processes, do not exhibit similar behavior.

Follow-on studies reinforce the role of attention shocks. Seasholes and Wu (2007) show that price-limit-hit stocks in China, a prominent attention event, experience large inflows from individual investors who had not previously held these securities. Prices quickly mean-revert, consistent with temporary overreaction by retail traders. Greenwood and Nagel (2009) find that younger mutual fund managers, those more likely to represent less experienced investor bases, exhibit trend-chasing behavior during the technology bubble, leading to poor performance.

More recent research focuses on the environment in which retail trades occur. Ozik, Sadka, and Shen (2021) document that, during the COVID-19 lockdown, Robinhood investors disproportionately bought stocks that received intense pandemic-related media coverage. Barber et al. (2022) show that Robinhood's platform design, emphasizing "Top Movers" lists and simplified interfaces, reinforces attention-driven buying and leads to localized herding in salient stocks. Their findings highlight that platform architecture can magnify behavioral biases and shape the kinds of stocks retail investors disproportionately hold.

Yet the literature also contains counter-evidence. Welch (2022) shows that the aggregate Robinhood portfolio performs surprisingly well on a risk-adjusted basis, suggesting that retail investors may not behave uniformly as noise traders. Eaton et al. (2021) identify cases of informed retail trading, such as ahead of certain news.

A key limitation persists as most research uses proxies such as abnormal volume, searches, and app surges that capture transient attention rather than persistent ownership. This raises questions about whether systematic retail participation, tracked via weekly holdings, drives mispricing similarly or interacts with anomalies like MAX. Our study leverages Robinhood holdings data to

distinguish sustained retail involvement from short-lived bursts of activity, thereby addressing this gap in the context of evolving views on retail investor sophistication.

2.2. Lottery-Like Stocks and the Pricing of Extreme Returns

The second relevant literature examines investors' preference for lottery-like payoffs, assets offering a small probability of extremely large returns. This preference is well grounded in psychology and prospect theory. Investors overweight small probabilities (Tversky and Kahneman, 1992), leading them to overvalue assets with positively skewed payoff distributions. Barberis and Huang (2008) formalize this mechanism, showing that skewness-seeking behavior can produce overpriced securities and negative expected returns.

The empirical literature identifies several proxies capturing the lottery-like characteristics of stocks. Kumar (2009) shows that low-priced stocks with high idiosyncratic volatility and high positive skewness are disproportionately held by individual investors and deliver poor subsequent returns. Boyer et al. (2010) propose expected idiosyncratic skewness as an alternative measure and show that stocks with higher predicted skewness underperform. Conrad et al. (2014) emphasize jackpot probability and find that stocks with high ex ante probabilities of extreme payoffs earn unusually low returns. Recent work has documented a lottery factor across anomalies and extensions to high-retail markets, such as China, where skewness premiums are strengthened.

Among these proxies, the MAX effect introduced by Bali, Cakici, and Whitelaw (2011) is particularly influential. Their results show a robust and economically large negative relation between MAX and future returns. Investors appear to overreact to large positive daily shocks, driving up prices and leading to underperformance in the following month.

Subsequent research refines this understanding. Nguyen and Truong (2018) show that the MAX effect is weaker when extreme returns coincide with earnings announcements, suggesting that part of MAX reflects information rather than speculation. Fong and Toh (2014) show that MAX is stronger following high-sentiment episodes.

2.3. Retail Participation and Lottery Stocks: The Missing Link

These literatures converge on two patterns in which retail investors overweight speculative, positively skewed stocks and lottery-like assets, especially high-MAX stocks, tend to underperform. This has fostered a narrative that retail demand contributes to lottery mispricing. Yet this link is largely assumed, not tested, particularly with direct holdings data.

Methodologically, prior studies capture retail interest via attention proxies, extreme returns, volume spikes, or news events (Barber and Odean 2008; Barber et al. 2022), which reflect episodic herding rather than steady-state ownership. As a result, evidence on retail's role in anomalies like MAX is indirect and potentially confounded, since a MAX event itself is salient and may trigger the short-lived inflows captured by prior proxies but not necessarily sustained participation.

Several key assumptions thus remain unverified. Retail investors are presumed to systematically overprice lottery stocks, high retail ownership is assumed to amplify MAX underperformance, and retail investors are viewed as acting as the marginal pricer in speculative market segments.

This gap is particularly pronounced for MAX, often viewed as a signature of speculative demand yet unexamined conditional on retail holdings. Does MAX weaken in low-retail stocks or intensify in high-retail ones? Our study addresses this gap by leveraging weekly Robinhood holdings data to construct measures of systematic participation, net buying pressure, size- and price-adjusted abnormal ownership, and microstructure-residualized trading flows. Double-sorted portfolios on these measures and MAX test whether the anomaly varies with retail intensity, providing direct evidence on the presumed mechanism.

2.4. Hypothesis Development

The literature on retail investor behavior and the pricing of lottery-like stocks has developed along separate tracks, yet the two share a common foundation. Both often point to retail traders as a source of mispricing in speculative stocks. Studies of attention-driven trading show that sharp spikes in salience, extreme daily returns, unusual volume, or news coverage, draw in retail buyers who push prices up temporarily, only for those prices to fall back as the excitement fades (Barber and Odean 2008). This pattern has built a strong view that retail demand, especially when it looks impulsive and sentiment-fueled, helps explain why lottery-like stocks earn low returns over time.

That view, however, rests on evidence drawn almost entirely from short-lived bursts of activity. The proxies used in earlier work, abnormal trading volume on a single day, Google search spikes, or sudden jumps in Robinhood app rankings, pick up exactly those moments when a stock grabs widespread notice. They do not capture the quieter, week-to-week presence of retail investors in a stock such as how many continue to hold it, whether they are steadily adding shares, or whether the stock has become a regular part of their portfolios.

The holdings data used here offer a different lens. By tracking the number of Robinhood users holding each stock at the end of each week and adjusting these counts for mechanical effects related to firm size, stock price, and short-term market conditions, our measures isolate stocks in which retail investors maintain an unusually persistent presence. This is not the same as the transitory buying waves that follow a big one-day price jump. A stock may experience intense attention in one week and then quickly fade from retail portfolios. Conversely, a stock may remain quietly held in thousands of retail accounts week after week without ever appearing among top movers.

This difference in what is being measured opens the door to a different prediction about returns. When retail buying is driven by a sudden salient event, other investors can quickly recognize the temporary nature of the pressure and step in to supply liquidity, leading prices to revert. Persistent retail ownership, by contrast, reflects sustained participation rather than transitory attention shocks. Such steady demand does not generate the same immediate arbitrage incentives, particularly in stocks where trading is costly or limits to arbitrage are binding. As a result, prices

may adjust more gradually over subsequent weeks as retail investors continue to accumulate positions, allowing positive return predictability to emerge without immediate reversal.

Moreover, steady retail inflows create ongoing buying pressure in stocks that are often smaller and less liquid. Unlike a one-day surge that attracts immediate contrarian trading, persistent demand is harder to trade against profitably, allowing positive returns to build gradually rather than reverse.

For these reasons, stocks with higher systematic retail participation, measured through sustained abnormal ownership or cleaned weekly flows, may earn higher subsequent returns. This leads to the first hypothesis:

H1 (Retail Outperformance Hypothesis): Stocks with greater systematic retail participation earn higher future returns, in contrast to the negative returns typically associated with attention-driven retail trading.

The second hypothesis concerns the link most papers take for granted that retail demand is the main force behind the underperformance of high-MAX stocks. The MAX anomaly, a robust negative relation between a stock's highest daily return in a month (or week) and its subsequent return, is frequently interpreted as evidence of speculative overpricing driven by lottery-seeking investors (Bali, Cakici, and Whitelaw 2011). Because retail investors are thought to chase extreme returns most aggressively, the anomaly is often attributed, at least in part, to their buying behavior.

Yet the evidence for that attribution comes from the same attention-based proxies. A large daily return spike is itself a salient event that triggers the kinds of short-lived retail inflows captured by earlier studies. It does not necessarily follow that the baseline level of retail ownership in a stock, the degree to which retail investors steadily hold or accumulate it, drives or amplifies the later reversal. If the reversal after extreme returns stems primarily from the temporary excitement around the spike itself, then the magnitude of the MAX effect should not vary strongly with the degree of ongoing retail participation.

Double-sorted portfolios provide a direct way to test this. If retail ownership were the marginal driver, high-MAX stocks should underperform most dramatically when systematic retail participation is elevated. Finding instead that the MAX spread remains similar across the retail distribution would indicate that the two phenomena operate largely independently.

This leads to the second hypothesis:

H2 (Retail-MAX Independence Hypothesis): The negative relation between MAX and future returns is similar across levels of systematic retail participation.

The third hypothesis asks whether this independence holds up when we separate environments that naturally affect both retail activity and extreme returns. Earnings announcements, for example, produce legitimate large price moves and draw genuine attention, potentially influencing both MAX and retail measures. Periods of high market sentiment or low liquidity can also mechanically boost retail flows and extreme daily returns. If the lack of interaction between systematic retail

participation and MAX is genuine rather than driven by these conditions, the pattern should survive controls for earnings windows, firm size, book-to-market, liquidity, and sentiment.

This leads to the third hypothesis:

H3 (Robustness Hypothesis): The relationship between systematic retail participation and the MAX effect remains stable after accounting for earnings announcements, liquidity conditions, firm characteristics, and market sentiment.

3. Data Description

Retail investor activity is measured using holdings data originally obtained from Robintrack.net, a platform that continuously scraped Robinhood's publicly available API. The API reported, approximately once per hour, the number of Robinhood users holding at least one share of each stock. Following the data-cleaning procedures outlined in Welch (2022), repeated intrahour observations and entries where the number of users did not change are removed, producing a large and informative panel of roughly twelve million ticker-hour observations. For empirical analysis, the dataset is collapsed to the stock-day level by retaining the final UTC-stamped observation of each trading day, yielding approximately 5.8 million stock-day observations across 802 trading days.

These observations are merged with CRSP through standard identifier matching. Out of 8,560 tickers reported on Robinhood during the sample period, 8,387 are successfully linked to CRSP, and 3,834 correspond to common equity securities (CRSP share codes 10 or 11). A small number of tickers never appear in the data, either due to Robinhood API limitations or early versions of the Robintrack scraper that failed to record certain dual-class shares before early 2020. In addition, several days contain missing entries caused by platform-wide outages (notably on March 2–3 and March 9, 2020) or temporary failures in the scraping script. Because the analysis uses weekly aggregations and weekly changes, these isolated gaps do not materially affect the construction of retail participation measures.

From the daily series of Robinhood user counts, we construct several weekly measures of retail participation designed to capture systematic, persistent involvement rather than short-lived, attention-induced trading bursts. The first measure is NetRetail (Sharesout), defined as the weekly change in the number of Robinhood holders scaled by shares outstanding. This variable reflects changes in retail ownership that are economically comparable across firms of different sizes and captures net buying pressure by platform users. The second measure, AbnRetail(P&S), captures abnormal retail ownership relative to firm size and price. It is obtained by extracting the portion of log user holdings unexplained by cross-sectional variation in firm size and stock price, identifying stocks that retail investors hold more heavily than predicted by their basic characteristics.

A third measure, AbnRetailFlow(P&S), captures abnormal weekly retail flow after removing the influence of size and price on changes in retail ownership. This isolates unexpected increases or

decreases in retail holdings that are not mechanically related to firm scale or trading constraints. Finally, AbnRetailFlow(Micro) incorporates a wider set of lagged microstructure characteristics, including volatility, turnover, and short-term returns.

Unlike proxies based on trading volume, search activity, or social media trends, these measures rely directly on holdings data and therefore capture the underlying retail investor base rather than episodic bursts of speculative interest. Because the defining feature of the Robinhood platform is its concentration of young, low-wealth, and speculative traders, these measures offer a behaviorally relevant view of the retail segment of the market.

Daily security-level data, including returns, prices, shares outstanding, and trading volume, are obtained from the CRSP daily files. Weekly stock returns are computed by compounding daily returns from Monday through Friday. To characterize the lottery-like features of individual stocks, several measures commonly used in the literature are constructed. The primary measure is MAX, defined as the highest daily return within a given week, following the approach of Bali, Cakici, and Whitelaw (2011). The analysis centers on MAX because it is both the theoretically most relevant lottery proxy and the one most compatible with the weekly horizon of the retail data.

A series of standard weekly firm characteristics is also incorporated as control variables. These include market capitalization (Size), book-to-market ratios derived from Compustat accounting data, turnover as a proxy for trading activity, short-term reversal based on prior-week returns, and intermediate momentum computed over multi-month horizons.

Quarterly accounting data from Compustat provides book values used to compute book-to-market ratios. Compustat's Earnings Announcement (EA) file is used to classify each stock-week as either an EA week or a non-EA (NOEA) week. Nguyen and Truong (2018) show that the MAX effect weakens when extreme returns occur close to earnings announcements, suggesting that some MAX realizations reflect information rather than speculative trading. While their analysis focuses on a narrow five-day announcement window, our weekly setting motivates a broader but still local classification. Specifically, we define EA weeks as the week of the earnings announcement as well as the adjacent week before and after. This captures the period over which earnings-related information is incorporated into prices at the weekly frequency. Separating EA and NOEA periods enables a comparison of whether the MAX anomaly and its interaction with retail participation are driven by informative events or by speculative trading.

To control for standard-priced risks, we use the Fama–French six-factor model (FF6) from Ken French's data library, which includes market, size, value, investment, profitability, and momentum factors, and the q-factor model (Hou, Xue & Zhang 2015) from Lu Zhang's website. All performance evaluations, including portfolio alphas and risk-adjusted return spreads, are computed relative to both benchmark models to ensure robustness.

The final weekly panel contains 393,693 stock–week observations and serves as the basis for all empirical tests. From this structure, we construct single-sorted portfolios on retail participation

and MAX, double-sorted portfolios that jointly sort on these dimensions, and a set of robustness portfolios partitioned by earnings announcements, firm size, and book-to-market characteristics.

4. Methodology

This section describes how the variables are constructed, how the portfolios are formed, how returns are evaluated relative to standard asset-pricing models, and how the interaction between retail participation and the MAX anomaly is tested.

4.1 Variable Construction

Most variables used in the analysis are defined in Appendix A. In this section, we formally describe the construction of the key variables, with particular emphasis on the retail participation measures and their underlying economic rationale. Because this study lies in distinguishing systematic retail involvement from attention-driven spikes, the construction of these retail variables requires justification and transparency.

4.1.1. Weekly Returns and Lottery Measures

Weekly returns are computed by compounding CRSP daily returns from Monday through Friday. The main proxy for lottery-like payoffs is MAX, defined as the highest daily return within the week:

$$MAX_{i,t} = \max \{r_{i,t}^{(Mon)}, \dots, r_{i,t}^{(Fri)}\}.$$

MAX captures extreme right-tail realizations, which the literature associates with speculative demand and mispricing. Because MAX is inherently a high-frequency measure, it aligns naturally with the weekly panel used in this study.

4.1.2. Retail Net Buying (Share-Adjusted)

To capture systematic retail buying pressure at the weekly horizon, we construct a share-adjusted measure of net retail inflows. Let $Holders_{i,w}$ denote the number of distinct Robinhood user accounts holding at least one share of stock i at the end of week w . This variable is obtained from the final UTC-stamped Robintrack observation of each trading day and averaged to the weekly level. Let $SharesOut_{i,w}$ represent the total number of shares outstanding for stock i in CRSP at the end of week w . Because raw holder counts tend to scale mechanically with firm size, the ratio $Holders_{i,w}$ over $SharesOut_{i,w}$ provides a normalized measure of the intensity of retail ownership. Weekly retail net buying is defined as the week-over-week change in this share-adjusted ownership measure:

$$NetRetail(Sharesout)_{i,w} = \left(\frac{Holders_{i,w}}{SharesOut_{i,w}} \right) - \left(\frac{Holders_{i,w-1}}{SharesOut_{i,w-1}} \right).$$

This variable captures persistent inflows and outflows of retail investors into each stock in a manner that is comparable across firms of different sizes. A positive value reflects net retail accumulation during the week, while a negative value indicates net selling or declining participation. By adjusting for shares outstanding, NetRetail (Sharesout) ensures that a change of, for example, 1,000 new users in a small-cap firm carries proportionally more information than the same change in a mega-cap firm.

4.1.3. Abnormal Retail Ownership (Size–Price Adjusted)

Raw counts of Robinhood holders vary systematically with firm characteristics, particularly size and price. Larger firms naturally attract more users due to visibility, index inclusion, and broader analyst coverage, while low-priced stocks tend to attract disproportionately high numbers of retail traders with lottery-like preferences. To distinguish true retail concentration from these mechanical effects, we construct an abnormal ownership measure that isolates the component of retail holdings unexplained by size and price fundamentals.

Let $Holders_{i,t}$ denote the number of Robinhood users holding stock i at the end of trading day t . Let $Mkt_Value_{i,t}$ represent the market value of equity (shares outstanding $\times$ price), and $Price_{i,t}$ denote the closing share price from CRSP. We estimate each day's cross-sectional relationship between retail holdings and these two core variables:

$$\log(Holders_{i,t}) = \alpha_t + \beta_1 \log(Mkt_Value_{i,t}) + \beta_2 \log(Price_{i,t}) + \varepsilon_{i,t}.$$

This model extracts the expected level of retail ownership for a stock of a given size and price on day t . The abnormal retail ownership is defined as the average daily residual in week t :

$$AbnRetail(P\&S)_{i,w} = \frac{1}{D(w)} \sum_{t \in D(w)} \hat{\varepsilon}_{i,t}.$$

where $D(w)$ is the set of trading days in week w , and the number of trading days that week. A positive residual indicates that stock i is held much more heavily by retail investors than comparable firms of similar size and price, while a negative residual indicates unusually low retail presence. This measure reveals where retail traders are disproportionately concentrated, independent of mechanical relationships rooted in firm fundamentals. Because the regression is estimated each day separately, $AbnRetail(P\&S)$ captures cross-sectional anomalies rather than time-series drift in platform size or trading activity.

4.1.4. Abnormal Retail Flow (Size–Price Adjusted)

This measure focuses on isolating unexpected changes in retail investor activity after controlling for mechanical variation driven by firm size and stock price. To capture the surprising component of retail inflows, rather than flows that simply scale with firm visibility, we begin by computing the weekly retail net buying, $\Delta \left(\frac{Holders_{i,w}}{SharesOut_{i,w}} \right)$. To purge predictable variation, we estimate the following weekly cross-sectional regression:

$$\Delta \left(\frac{\text{Holders}_{i,w}}{\text{SharesOut}_{i,w}} \right) = \gamma_t + \delta_1 \log (\text{Mkt_Value}_{i,w-1}) + \delta_2 \log (\text{Price}_{i,w-1}) + u_{i,t},$$

where $\text{Mkt_Value}_{i,w-1}$ is the firm's market capitalization in the prior week and $\text{Price}_{i,w-1}$ is its lagged stock price. The fitted residual, represents the abnormal retail flow:

$$\text{AbnRetailFlow(P\&S)}_{i,w} = \hat{u}_{i,w},$$

The component of weekly retail inflow that cannot be explained by firm size or price. Positive values correspond to unusually strong inflows relative to what the stock's fundamentals would predict, while negative values denote unexpectedly weak inflows or retail exits.

4.1.4. Abnormal Retail Flow (Microstructure-Adjusted):

The final retail participation measure further isolates unexpected retail inflows by removing not only the predictable effects of firm size and price, but also the influence of short-horizon market microstructure conditions. Weekly changes in retail holdings often covary mechanically with volatility, turnover, or short-term returns, factors that may prompt retail investors to rebalance or react to transitory news. To ensure that the measure reflects abnormal retail trading, we begin with the same weekly change in share-adjusted holders, $\Delta \left(\frac{\text{Holders}_{i,w}}{\text{SharesOut}_{i,w}} \right)$, we then regress this weekly change on a richer set of lagged firm characteristics that capture microstructure-driven trading patterns:

$$\Delta \left(\frac{\text{Holders}_{i,w}}{\text{SharesOut}_{i,w}} \right) = \eta_t + \theta' X_{i,w-1} + \omega_{i,w},$$

where the vector of lagged controls is defined as

$$X_{i,w-1} = [\log (\text{Mkt_Value}), \log (\text{Price}), \text{Volatility}_{21d}, \text{Turnover}, \text{PastRet}_{21d}]_{i,w-1}.$$

Here, Volatility_{21d} is the 21-day realized return volatility, Turnover is share turnover, and PastRet_{21d} captures short-horizon momentum or reversal forces. Each variable is lagged by one week to ensure that the regression explains predictable flow components rather than incorporating contemporaneous reactions.

The microstructure-adjusted abnormal retail flow is then given by the regression residual:

$$\text{AbnRetailFlow(Micro)}_{i,w} = \hat{\omega}_{i,w}.$$

Because this specification isolates the systematic microstructure component, $\text{AbnRetailFlow(Micro)}$ serves as an indicator of idiosyncratic shifts in retail sentiment and is therefore especially informative for testing whether retail activity interacts with the MAX anomaly.

4.2 Portfolio Formation

The main tests rely on portfolio sorts, which are implemented at the end of each week using information available at that time, with portfolios held over the subsequent week. When value-weighted returns are reported, weights are based on lagged market capitalization at the formation date. Equal-weighted and value-weighted results are shown throughout to distinguish average stock behavior from dollar-weighted outcomes.

The analysis begins with single-sorted portfolios that establish two benchmark patterns. First, stocks are sorted into quintiles based on the previous week's MAX, and the difference in next-week returns between the highest-MAX and lowest-MAX portfolios is computed. This spread summarizes the presence and magnitude of the MAX anomaly in the weekly data. Second, separate single sorts are performed on each of the four retail participation measures. Sorting stocks into quintiles and examining the return differences between high- and low-retail portfolios.

The primary tests use dependent double sorts on retail activity and MAX. At the end of each week, stocks are first sorted into five quintiles based on a given retail participation measure (for example, AbnRetail(P&S)). Within each retail quintile, stocks are then sorted again into five sub-portfolios based on MAX. This procedure yields a 5×5 set of portfolios that jointly capture the distribution of retail participation and lottery intensity. For each retail quintile, the High-MAX minus Low-MAX return spread is computed for the following week. Comparing these spreads across retail quintiles reveals whether the MAX anomaly becomes stronger, weaker, or remains unchanged as retail participation increases. The same double-sort procedure is repeated for each of the four retail measures and for both equal- and value-weighted returns.

To assess whether return patterns are explained by standard risk factors, all portfolio returns and High-minus-Low spreads are evaluated relative to two benchmark asset-pricing models, the Fama–French six-factor model and the q-factor model of Hou, Xue, and Zhang (2015). For each portfolio, time-series regressions of weekly excess returns on the factor sets are estimated, and risk-adjusted alphas are reported together with Newey and West (1987) t-statistics that correct for heteroskedasticity and autocorrelation.

4.3 Cross-Sectional Regression Tests

To complement the portfolio-based evidence, we conduct cross-sectional return tests using the Fama–MacBeth (1973) regression framework. These regressions allow us to examine whether retail participation predicts returns after conditioning on MAX and whether retail activity systematically alters the pricing of lottery-like characteristics in the cross section.

Specifically, for each week t , we estimate the following cross-sectional regression at the stock level:

$$R_{i,t+1} = \alpha_t + \beta_{1,t} \text{MAX}_{i,t} + \beta_{2,t} \text{Retail}_{i,t} + \beta_{3,t} (\text{MAX}_{i,t} \times \text{Retail}_{i,t}) + \gamma_t' \mathbf{M}_{i,t} + \varepsilon_{i,t+1},$$

where $R_{i,t+1}$ is the excess return of stock i in week $t + 1$, $MAX_{i,t}$ is the maximum daily return within week t , and $Retail_{i,t}$ denotes one of the four retail participation measures described in Section 4.1. The interaction term $MAX_{i,t} \times Retail_{i,t}$ captures whether the pricing of extreme returns varies with the intensity of retail participation.

The vector $M_{i,t}$ includes standard firm characteristics used in the anomaly and lottery literature, such as firm size, book-to-market ratio, market beta, turnover, short-term reversal, intermediate-horizon momentum, Idiosyncratic Volatility (IVOL), and Idiosyncratic Skewness (ISKEW). All control variables are measured at week t and are defined in Appendix A. Including these controls ensures that the estimated coefficients on MAX and retail participation are not driven by known cross-sectional return predictors.

The weekly slope coefficients are averaged over time, and statistical inference is based on the time-series standard errors of the estimated coefficients, with Newey–West adjustments to account for potential autocorrelation in the coefficient estimates.

This framework provides a direct test of whether retail participation strengthens or weakens the MAX anomaly. A significantly negative average coefficient on $MAX_{i,t}$ confirms the presence of the MAX effect in weekly data, while a significantly positive coefficient on $Retail_{i,t}$ indicates that systematic retail participation predicts higher future returns. Most importantly, an insignificant coefficient on the interaction term $MAX_{i,t} \times Retail_{i,t}$ implies that the negative relation between MAX and future returns is largely invariant to the level of retail participation.

4.4 Robustness and Subsample Analyses

Several methodological extensions are implemented to verify that the results are not driven by particular subsets of the data or by obvious alternative explanations. First, the entire portfolio-sorting procedure is repeated separately for earnings-announcement weeks and non-announcement weeks. Weeks falling within three weeks of a firm's earnings release are labeled EA weeks, while all other weeks are classified as NOEA weeks. This split addresses the concern that very high MAX realizations around earnings announcements may reflect information-driven price discovery rather than speculative trading. Comparing the retail–MAX interaction in EA and NOEA weeks reveals whether the absence of a strong retail channel in the baseline results is robust to this information environment.

Second, to verify that the documented patterns are not driven mechanically by firm size or valuation, we conduct triple-sorted portfolio tests that jointly condition on firm size or book-to-market, retail participation, and MAX. At the beginning of each week, stocks are first sorted into terciles based on firm size or book-to-market using NYSE breakpoints to ensure comparability with the asset-pricing literature. Within each size or book-to-market tercile, stocks are then sorted into terciles based on the relevant retail participation measure. Finally, within each retail participation bin, stocks are further sorted into quintiles based on MAX. This procedure yields a $3 \times 3 \times 5$ set of portfolios and ensures sufficient cross-sectional variation within each cell. In

particular, we require that each resulting portfolio contain at least 20 stocks, which mitigates concerns about small portfolio sizes and ensures that return estimates are not driven by a small number of extreme observations.

5. Empirical Results

This section presents the empirical evidence on the relationship between systematic retail participation and the MAX anomaly. The analysis is structured to move from descriptive evidence to tests that isolate whether retail participation explains, strengthens, or is independent of lottery-like return patterns captured by MAX.

Two empirical patterns motivate the analysis, the MAX anomaly and the widely held view that retail investors prefer lottery-like stocks and may contribute to lottery-related mispricing. These observations suggest that if retail investors over-demand lottery-like payoffs, the MAX anomaly should be strongest among stocks with high retail participation. This section examines that hypothesis directly.

The empirical strategy proceeds in four steps. First, we characterize the cross-section of retail participation by documenting how abnormal retail ownership and flows relate to firm characteristics, trading activity, and lottery-like features. Second, we establish two baseline pricing facts at the weekly horizon (i) the MAX anomaly is strongly negative, and (ii) systematic retail participation, measured using holdings- and flow-based proxies, predicts positive, not negative, future returns. We then show that this retail return premium persists and accumulates over multi-week horizons, ruling out short-lived attention effects or microstructure-driven reversals. Third, we examine whether retail participation interacts with the MAX anomaly. Using a series of double-sorted portfolios and Fama–MacBeth cross-sectional regressions, we test whether the negative MAX premium varies with the level of retail ownership or trading intensity. Finally, we evaluate whether the independence between retail participation and MAX persists during earnings-announcement and non-announcement periods, and whether it varies across firm size and book-to-market groups.

5.1. Retail Participation, Lottery Characteristics, and the Cross-Section

We begin by examining how systematic retail participation relates to firm characteristics and lottery-like attributes in the cross-section. This validates that our retail participation measures capture the type of speculative demand emphasized in the lottery literature and establishes a descriptive benchmark that allows us to later distinguish retail sorting into lottery-like stocks from retail-related mispricing.

Table 1 reports summary statistics for quintile portfolios formed by sorting stocks on abnormal retail ownership adjusted for size and price, $AbnRetail(P\&S)$. Portfolios are formed weekly from May 2018 to August 2020, and the table reports time-series averages of within-week median firm characteristics.

A monotonic relation emerges between abnormal retail ownership and lottery-like stock features. The maximum daily return (MAX) increases steadily from 2.58% in the lowest retail quintile to 5.55% in the highest, more than doubling across the distribution. This pattern directly confirms that retail investors disproportionately hold stocks exhibiting extreme right-tail realizations, a defining feature of lottery-type payoffs emphasized by Bali, Cakici, and Whitelaw (2011).

Other lottery characteristics display even stronger patterns. Idiosyncratic volatility (IVOL) rises from 1.62 to 4.04, while idiosyncratic skewness (ISKEW) increases dramatically from 7.69 in the lowest retail quintile to 20.54 in the highest. The composite lottery index (Zscore), which aggregates MAX, IVOL, and ISKEW, moves monotonically from -0.50 to 0.38 . These patterns indicate that stocks with high retail participation offer payoff distributions that are substantially more volatile, more skewed, and more lottery-like along all standard dimensions used in the literature.

Retail participation is also strongly associated with trading intensity and firm characteristics. Share turnover increases from 5.74 in the lowest retail quintile to 65.02 in the highest, indicating that retail-intensive stocks are subject to exceptionally high trading activity and attention. Market beta rises from 0.74 to 1.19, suggesting that high-retail stocks exhibit greater systematic risk exposure.

In terms of fundamentals, high-retail stocks tend to be somewhat larger in this sample, with log size increasing from 5.99 to 7.45, and display meaningful variation in book-to-market ratios. While the lowest and highest retail quintiles exhibit similar average BTM ratios (0.97 and 1.01, respectively), intermediate quintiles show lower values, consistent with retail investors' preference for growth-oriented and speculative firms rather than value stocks.

Return characteristics provide additional evidence on the nature of systematic retail participation. Past momentum (MOM), measured as cumulative returns over the prior eleven weeks excluding the most recent week, increases sharply from -2.25% in the lowest retail quintile to 7.99% in the highest. Formation-week returns (REV) follow the same pattern, rising from -0.10% to 0.77% . These patterns indicate that retail investors disproportionately hold stocks with strong recent performance and high returns.

Importantly, these return-based characteristics suggest that AbnRetail(P&S) captures persistent retail engagement rather than short-lived attention shocks. Retail investors appear to concentrate in stocks that are already volatile, actively traded, and exhibiting strong recent performance, consistent with persistent speculative interest rather than transient noise trading.

To ensure that these findings are not specific to abnormal retail ownership, we replicate the same summary-statistics analysis for three additional retail participation measures: NetRetail(SharesOut), AbnRetailFlow(P&S), and AbnRetailFlow(Micro). Appendix B Tables B1–B3 show remarkably similar monotonic patterns across all four proxies. Regardless of whether retail activity is measured using ownership levels or flow-based measures, higher retail participation is consistently associated with higher MAX, greater volatility, stronger right-skewness, and substantially higher trading intensity.

5.2. Single-Sorted Portfolio Evidence

This section evaluates the predictive content of lottery characteristics and retail participation using single-sorted portfolios. We begin by confirming the presence of the MAX anomaly at the weekly horizon. We then examine whether systematic retail participation predicts future returns and assess the economic magnitude of this effect relative to MAX.

5.2.1. The MAX Anomaly at the Weekly Horizon

First column of Table 2 reports portfolio returns for stocks sorted into quintiles based on MAX, defined as the maximum daily return realized during the previous week. Consistent with the established lottery literature, the MAX anomaly is present in weekly data. The equal-weighted return spread between the highest-MAX and lowest-MAX portfolios is -0.27% per week, with a corresponding t-statistic of -0.87 . Risk adjustment using the Fama–French six-factor model and the q-factor model yields nearly identical results.

Value-weighted results reinforce this conclusion. The high-minus-low MAX spread remains negative at -0.19% per week, suggesting that the anomaly is not confined to micro-cap stocks or driven solely by equal-weighting. Findings confirm that stocks exhibiting extreme right-tail daily returns continue to underperform in the cross-section, even at a weekly horizon, in line with Bali, Cakici, and Whitelaw (2011) and related work.

5.2.2. Retail Participation Predicts Positive Returns

In contrast to MAX, columns 2 to 5 of Table 2 show that all four retail participation measures predict positive future returns. When stocks are sorted on abnormal retail ownership, $\text{AbnRetail(P\&S)}$, the equal-weighted high-minus-low return spread is 0.49% per week, with a t-statistic of 2.65. Abnormal retail flow measures deliver similar results, 0.45% per week for $\text{AbnRetailFlow(P\&S)}$ ($t = 2.10$) and 0.33% per week for $\text{AbnRetailFlow(Micro)}$ ($t = 2.21$). $\text{NetRetail(SharesOut)}$ also exhibits positive return predictability, though with somewhat weaker statistical significance.

These patterns persist under value-weighting, with high-retail portfolios continuing to outperform low-retail portfolios across all four measures. These results show that the retail participation premium is not driven by small, illiquid, or extreme lottery stocks. Instead, the evidence suggests that persistent retail involvement is associated with higher subsequent returns across a broad segment of the cross-section.

5.2.3. Why Does Retail Participation Earn Positive Returns?

The positive pricing of retail participation is salient because it directly contradicts the conventional view in which retail investors are viewed primarily as noise traders who overpay for lottery-like stocks. A distinction in our setting is that our retail measures capture systematic and persistent participation, not transient attention shocks. $\text{AbnRetail(P\&S)}$ reflects persistent deviations in retail

ownership relative to firm fundamentals, while flow-based measures capture ongoing net buying rather than short-lived bursts of demand.

From this perspective, persistent retail participation is more appropriately interpreted as gradual information dissemination or durable investor engagement rather than speculative overreaction. Retail investors may accumulate positions over time in firms they understand, follow closely, or repeatedly trade, generating a slow-moving demand component that is not immediately incorporated into prices. This mechanism is consistent with theories of limited attention and slow information diffusion, in which prices adjust gradually as information spreads across investor groups. (Hong and Stein, 1999; Sims, 2003; DellaVigna and Pollet, 2009; Da, Guren, and Warachka, 2014).

Comparing magnitudes highlights the economic relevance of these findings. The weekly MAX anomaly ranges from -0.19% to -0.27% , depending on weighting, while the retail participation premium ranges from approximately $+0.33\%$ to $+0.49\%$ per week in equal-weighted portfolios. This comparison underscores a key point that retail participation is not a second-order effect relative to lottery anomalies. Instead, it represents a quantitatively important and economically meaningful return predictor that operates in the opposite direction.

The evidence in Table 3 further supports this interpretation, showing that when portfolios sorted on AbnRetail(P&S) are held for longer horizons, return differentials not only persist but also accumulate. The equal-weighted high-minus-low spread grows from 0.49% at the one-week horizon to 5.13% over eleven weeks, with t-statistics remaining around 2.6 – 2.7 throughout. Value-weighted results exhibit the same monotonic pattern. These dynamics are difficult to reconcile with temporary mispricing or microstructure effects, which would be expected to reverse quickly. Instead, they point to a slow-moving return component associated with persistent retail demand.

5.3. Double-Sorted Portfolios: Retail Participation $\times$ MAX

The coexistence of a negative MAX premium and a positive retail participation premium raises a central question whether retail participation amplifies the MAX anomaly. A large body of the lottery literature implicitly assumes that speculative retail demand is the primary force behind the underperformance of lottery-like stocks. If this view is correct, the MAX effect should be strongest precisely where retail participation is highest. This section evaluates that hypothesis using double-sorted portfolios that jointly condition on retail participation and MAX.

5.3.1. Abnormal Retail Ownership $\times$ MAX: Full Double Sort

Table 4 reports portfolio returns from double sorts in which stocks are first sorted into quintiles based on abnormal retail ownership, AbnRetail(P&S), measured at the end of the previous week, and then further sorted into quintiles based on MAX. Panel A reports equal-weighted results, while Panel B reports value-weighted results. Within each retail quintile, the final columns report the high-minus-low MAX spread in raw returns and risk-adjusted alphas.

The analysis yields two key findings. First, the MAX anomaly is present within every retail ownership quintile. In the equal-weighted results (Panel A), the high-minus-low MAX return spread is negative in all five AbnRetail(P&S) quintiles, ranging from -0.26% to -0.62% per week. The corresponding FF6 and q-factor alphas are similarly negative and statistically meaningful. Importantly, high-MAX stocks underperform low-MAX stocks not only among low-retail firms but also among firms with the highest levels of abnormal retail ownership.

The value-weighted results in Panel B reinforce this conclusion. Although the magnitudes are somewhat smaller, as is typical for MAX-related anomalies, the high-minus-low MAX spreads remain negative across all retail quintiles.

Second, the magnitude of the MAX effect does not increase with retail participation. If retail investors were the marginal force behind lottery mispricing, one would expect the MAX anomaly to be most pronounced in the highest retail quintiles. Table 4 provides no support for this prediction. The MAX spread in the highest AbnRetail(P&S) quintile is economically similar to, and in some cases smaller than, that in the lowest quintile. There is no monotonic pattern across retail ownership groups, and differences across quintiles are modest relative to their standard errors.

5.3.2. MAX Spreads across Retail Participation Measures

Table 5 generalizes the analysis by repeating the double-sort procedure using all four retail participation proxies: NetRetail(SharesOut), AbnRetail(P&S), AbnRetailFlow(P&S), and AbnRetailFlow(Micro). For expositional clarity, the table reports only the high-minus-low MAX return spreads and corresponding FF6 alphas within each retail quintile. Full portfolio-level results are reported in Appendix Tables B4–B6.

The equal-weighted results in Panel A show a remarkably consistent pattern. Across all four retail measures and all five retail quintiles, the MAX spread is negative. For example, conditioning on AbnRetail(P&S), the MAX spread ranges from -0.26% to -0.62% per week. Conditioning on flow-based measures yields nearly identical magnitudes. Net retail buying produces slightly noisier estimates, but the sign remains uniformly negative.

Panel B reports value-weighted results and delivers the same qualitative message. MAX underperformance persists across retail quintiles, with spreads typically between -0.11% and -0.40% per week. Once again, there is no evidence that the anomaly intensifies in retail-heavy stocks.

Appendix Tables B4–B6 confirm that these findings are not driven by selective reporting. When examining the full matrix of portfolio returns, high-MAX portfolios consistently underperform low-MAX portfolios within each retail group, while retail participation itself continues to load positively on returns.

While retail investors do concentrate on volatile and skewed stocks (as shown in Table 1), their sustained participation does not strengthen lottery-related underperformance. Instead, the MAX anomaly appears to operate independently of who holds the stock.

This distinction is important because MAX captures realized extreme payoffs and reflects properties of the return distribution, whereas retail participation captures persistent ownership and trading behavior. The double-sort results demonstrate that these two objects are not interchangeable. High-MAX stocks underperform regardless of whether retail ownership is low or high, and retail-heavy stocks do not exhibit disproportionately worse MAX penalties.

From an economic perspective, the double-sorted results also illustrate how MAX compares to retail participation in shaping returns. In single-sorted portfolios, the MAX anomaly is approximately -0.2% to -0.3% per week, while the retail participation premium is roughly $+0.3\%$ to $+0.5\%$ per week. The double-sorted evidence shows that these magnitudes remain largely unchanged when conditioning on the other variable.

5.4. Fama-Macbeth cross-sectional regressions

This section complements the portfolio evidence by examining the joint pricing of retail participation and the MAX characteristic in a Fama–MacBeth (1973) cross-sectional regression framework.

Table 6 reports results from weekly regressions of one-week-ahead stock excess returns on abnormal retail ownership, MAX, and their interaction. In each week, returns in week $t + 1$ are regressed on variables measured at the end of week t . The reported coefficients are time-series averages of the weekly slope estimates, with Newey–West adjusted t-statistics.

5.4.1. Retail Participation Earns a Positive and Robust Return Premium

Across all specifications, abnormal retail ownership, $\text{AbnRetail(P\&S)}$, loads positively and significantly on future returns. In the baseline specification, the estimated coefficient is 1.50 ($t = 2.51$), implying that a one-unit increase in abnormal retail ownership is associated with an increase of approximately 1.5% in expected weekly excess returns. This effect remains remarkably stable as controls are added incrementally. Even in the fully comprehensive specification, including beta, size, book-to-market, momentum, reversal, turnover, idiosyncratic volatility, and idiosyncratic skewness, the coefficient on $\text{AbnRetail(P\&S)}$ remains economically large ($\approx 1.5\text{--}1.6$) and statistically significant at standard levels.

This finding provides a direct answer to an important question left implicit in much of the lottery literature: why does systematic retail participation earn positive returns rather than negative ones? The regression evidence suggests that persistent retail ownership reflects more than transient attention or noise trading. Unlike short-lived retail order imbalances around news events, $\text{AbnRetail(P\&S)}$ captures sustained participation that is correlated with positive return continuation, even after controlling for momentum and reversal effects.

Economically, this pattern is consistent with interpretations in which retail investors contribute to gradual information diffusion, provide liquidity to constrained arbitrageurs, or disproportionately participate in stocks undergoing slow-moving repricing. In this sense, systematic retail participation behaves more like a persistent demand-side characteristic than a speculative overpricing signal.

MAX retains a strong, independent negative price of risk, with the baseline regression yielding a coefficient of -3.12 ($t = -4.86$) that increases slightly as additional controls are introduced, reaching -3.7 to -3.8 in the most comprehensive specifications. These estimates imply economically large underperformance, as a one-unit increase in MAX is associated with a reduction in expected weekly returns of roughly 3.5%.

Importantly, the incorporation of $\text{AbnRetail(P\&S)}$ has no weakening effect on the MAX coefficient. The MAX anomaly survives unchanged after controlling for retail participation, firm size, value, momentum, short-term reversal, liquidity, and higher-moment risks.

From an economic standpoint, the magnitude of the MAX effect exceeds that of retail participation. While $\text{AbnRetail(P\&S)}$ contributes positively to expected returns at roughly +1.5% per unit, MAX subtracts more than twice that amount.

Across all specifications, the interaction coefficient is statistically insignificant, and there is no evidence of a retail-MAX interaction. The coefficient estimates are positive, ranging from approximately 2.1 to 4.7, but t-statistics remain well below conventional significance thresholds, typically between 0.6 and 1.5. If retail investors were the marginal traders driving lottery mispricing, one would expect MAX underperformance to be more severe precisely in stocks with high retail ownership, yielding a significantly negative interaction term. The absence of such an effect provides strong evidence against this mechanism.

The regression results thus formalize what the double-sorted portfolios already suggested, MAX and retail participation affect returns additively rather than interactively. High-MAX stocks underperform regardless of whether retail ownership is low or high, and retail-heavy stocks earn positive returns regardless of their MAX exposure. Retail participation neither enhances nor mitigates the MAX anomaly.

Retail investors do hold lottery-like stocks, as shown in the summary statistics, but their sustained participation is not the mechanism that generates MAX-related underperformance. Instead, the MAX anomaly appears to reflect features of return distributions, such as extreme realizations and skewness, that are priced independently of ownership structure.

5.5. Conditioning on Earnings-Announcement and Non-Earnings Weeks

Earnings announcements are periods of concentrated information arrival, heightened attention, and high volatility, precisely the conditions under which both retail trading and lottery-like return realizations tend to intensify. Because MAX is mechanically more likely to increase sharply

around news events, and retail participation may also respond to salient information, it is important to verify that our earlier findings are not driven by earnings-announcement (EA) dynamics. This section, therefore, examines whether (i) the positive return premium to retail participation and (ii) the independence of the MAX anomaly from retail activity persists when the sample is split into EA and non-EA (NOEA) weeks.

5.5.1. Retail Participation and Future Returns in EA vs. NOEA Weeks

Table 7 reports single-sorted portfolio results based on retail participation measures, separately for EA and NOEA weeks. Two clear and economically important patterns emerge.

First, retail participation continues to predict higher future returns in both EA and NOEA windows, across all four retail measures. During EA weeks (Panel A), average returns increase monotonically across retail quintiles. For example, when sorting on $\text{AbnRetail(P\&S)}$, the lowest retail quintile earns -0.13% per week, while the highest quintile earns 0.41% , producing a high-minus-low spread of 0.54% per week ($t = 2.38$). Similar spreads appear for $\text{NetRetail(SharesOut)}$ (0.53%), $\text{AbnRetailFlow(P\&S)}$ (0.40%), and $\text{AbnRetailFlow(Micro)}$ (0.41%), with factor-adjusted alphas remaining positive and statistically meaningful under both FF6 and q-factor models.

Second, the retail return premium is at least as strong outside earnings windows. In NOEA weeks (Panel B), the high-minus-low spread for $\text{AbnRetail(P\&S)}$ rises to 0.57% per week ($t = 2.12$), while spreads for the flow-based measures reach 0.63% – 0.64% , with even stronger statistical significance in some cases. Importantly, the persistence of the retail premium in NOEA weeks indicates that it is not driven by short-lived earnings-related attention shocks or announcement-day reversals. Instead, the evidence supports the interpretation that systematic retail participation reflects a more stable return-relevant characteristic, consistent with gradual information dissemination, learning, or persistent demand rather than transient noise trading.

5.5.2. Retail Participation and the MAX Effect in EA vs. NOEA Weeks

Table 8 extends the EA/NOEA analysis to the interaction between retail participation and the MAX anomaly using double-sorted portfolios. For each retail measure, the table reports equal-weighted high-MAX minus low-MAX return spreads within retail quintiles, separately for EA and NOEA weeks.

The results are strikingly consistent with earlier sections. The MAX anomaly remains uniformly negative across retail participation quintiles in both EA and NOEA periods. During EA weeks (Panel A), MAX spreads are strongly negative in most retail quintiles, often larger in magnitude than in the full sample. For instance, when conditioning on $\text{AbnRetail(P\&S)}$, the low-retail quintile exhibits a MAX spread of -0.20% ($t = -2.33$), while the high-retail quintile shows a spread of -0.10% ($t = -1.19$). Similar negative patterns appear for $\text{NetRetail(SharesOut)}$ and the flow-based measures.

Importantly, there is no systematic monotonic relationship between retail intensity and the size of the MAX spread. High-retail portfolios do not display consistently larger MAX underperformance than low-retail portfolios, even in EA weeks when lottery-like returns and attention are most pronounced. In NOEA weeks (Panel B), the same pattern persists. MAX spreads remain negative across all retail quintiles, with magnitudes typically ranging from -0.35 percent to -0.70 percent per week, and again show no systematic strengthening in retail-heavy stocks.

Table B7 confirms that these conclusions are not driven by equal-weighting. The value-weighted EA and NOEA results show strongly negative MAX spreads across retail quintiles, often with even larger magnitudes during EA weeks (AbnRetail(P&S) spreads reaching -1.40% in the lowest retail quintile during EA periods). Yet, as before, retail participation does not change the MAX anomaly, and high-retail stocks do not exhibit disproportionately worse MAX-related underperformance.

This section presents two important findings. First, retail outperformance is present both when information is most salient (EA weeks) and when markets operate under normal conditions (NOEA weeks). Third, and most importantly, retail participation does not amplify the MAX effect, even during periods of intense attention and information flow.

5.6 Heterogeneity by Firm Characteristics

A natural concern is whether the documented interaction between retail participation and the MAX anomaly is limited to specific segments of the cross-section, such as microcap stocks or growth-oriented firms with lottery-like payoffs, or whether it reflects a more general phenomenon. To address this concern and strengthen the external validity of our findings, we examine heterogeneity in the retail–MAX relation across two fundamental firm characteristics, firm size and book-to-market ratios. We implement triple-sorted portfolio tests that condition on these characteristics while preserving the full retail participation and MAX sorting structure.

5.6.1 Firm Size Conditioning

We begin by examining whether the relationship between retail participation and the MAX anomaly varies systematically with firm size. This analysis addresses the concern that the observed retail–MAX patterns may be driven primarily by microcap stocks, where lottery features, retail trading, and limits to arbitrage are most pronounced. Tables 9 and 10 report equal-weighted triple-sorted portfolio results; Appendix Tables B8 and B9 present the corresponding value-weighted results.

Table 9 shows that the MAX anomaly is present across all size categories, small, medium, and big firms, although its magnitude declines monotonically with firm size. Among small stocks, the high-minus-low MAX spread is economically large and strongly negative, reaching -0.66% per week in the high-retail group, with corresponding FF6 and q-factor alphas of -1.15% and -1.22% , respectively. For medium-sized firms, MAX spreads remain negative but are smaller in magnitude, ranging between -0.18% and -0.50% per week, depending on retail intensity. Among large firms,

the MAX effect persists but is substantially weakened, with spreads generally between -0.14% and -0.43% per week.

This size pattern is consistent with the broader lottery literature, which emphasizes that MAX-related underperformance is strongest among smaller, more volatile, and more speculative stocks.

Retail participation plays a systematic role in shaping the magnitude of the MAX anomaly within size categories. Within each size tercile, high-retail stocks do not exhibit substantially more negative MAX spreads than low-retail stocks.

As shown in Table 10, within the small-size group, the MAX spread for high-retail stocks reaches -0.66% per week using $\text{AbnRetail(P\&S)}$, compared to -0.66% in the low-retail group. Similar pattern appears when retail participation is measured using $\text{NetRetail(SharesOut)}$, $\text{AbnRetailFlow(P\&S)}$, or $\text{AbnRetailFlow(Micro)}$.

Among medium-sized firms, the same qualitative pattern holds, although magnitudes are smaller. High-retail stocks exhibit MAX spreads between -0.18% and -0.69% per week, while low-retail stocks exhibit spreads closer to -0.31% to -0.59% .

For large firms, retail conditioning continues to matter, but the economic magnitude is substantially reduced. MAX spreads in large stocks are generally modest, often below -0.30% per week, and in some cases statistically indistinguishable from zero. Nevertheless, even among large firms, the MAX penalty tends to be negative in high-retail portfolios and in low-retail portfolios, indicating that retail participation does not significantly contribute to lottery-like mispricing.

These results yield three key findings. First, the MAX anomaly is strongest among small firms but remains present across the entire size distribution, confirming its importance. Second, retail participation systematically does not strengthen the MAX effect within each size group. Third, the retail-MAX interaction is not mechanically driven by firm size itself. Even after conditioning on size, retail intensity continues to explain meaningful variation in the severity of MAX underperformance.

Appendix Tables B8 and B9 present the value-weighted counterparts to Tables 9 and 10. Although value weighting attenuates magnitudes, as expected when larger firms receive more weight, the qualitative conclusions remain unchanged. MAX underperformance persists across size groups, retail participation does not magnify the MAX effect, and the retail premium remains visible beyond the smallest firms. This confirms that the size-conditioning results are not driven by extreme micro-cap observations.

5.6.2. Book-to-Market Conditioning

This subsection examines whether the relation between retail participation, the MAX anomaly, and future returns is driven by firms' valuation characteristics. Specifically, we replicate the triple-sort design of Section 5.6.1 while conditioning on book-to-market (BTM) ratios instead of firm size. This analysis addresses the concern that our earlier findings may reflect a growth-versus-value mechanism, given that lottery-like stocks are often associated with low BTM (growth) firms.

Tables 11 and 12 report equal-weighted results, while Appendix Tables B10 and B11 present value-weighted counterparts.

Table 11 reports triple-sorted portfolio returns by BTM terciles, abnormal retail ownership, and MAX. Across all three BTM groups, low, medium, and high, the MAX anomaly remains evident. Within each BTM tercile, portfolios with high MAX realizations earn substantially lower subsequent returns than their low-MAX counterparts, particularly among stocks with low retail participation.

For example, in the low-BTM panel (Panel A), the high-minus-low MAX return spread within the low-retail portfolio is -0.39% per week, with negative factor-adjusted alphas under both the Fama–French six-factor and q-factor models. Similar negative MAX spreads appear in the medium-BTM panel, where low-retail stocks exhibit a MAX return spread of -0.63% per week, accompanied by strongly negative alphas (-0.64% and -0.67%). Even among high-BTM (value) firms, the MAX anomaly persists, with low-retail stocks showing a high-minus-low MAX spread of -0.68% per week and statistically significant negative alphas. These patterns demonstrate that the MAX effect is not limited to growth stocks and is present across the valuation range.

Importantly, conditioning on book-to-market does not alter the key interaction result documented earlier that the MAX anomaly does not systematically intensify with retail participation. Within each BTM tercile, the magnitude of the MAX return spread does not increase monotonically as abnormal retail ownership rises. In several cases, the MAX penalty is actually weakened among high-retail portfolios. For instance, in the low-BTM panel, the MAX spread narrows from -0.39% in the low-retail group to -0.42% in the high-retail group. In the medium-BTM panel, the MAX spread is largest among low-retail stocks and smaller among high-retail stocks. Even in the high-BTM panel, where MAX spreads remain economically large, there is no evidence that retail participation strengthens the MAX anomaly. Instead, the MAX penalty appears largely independent of retail ownership intensity once valuation is held fixed.

Table 12 reinforces these conclusions by summarizing high-minus-low MAX return spreads across four alternative retail participation measures. Across all BTM terciles and retail proxies, MAX spreads remain negative on average, confirming the robustness of the MAX anomaly. However, consistent with Table 11, these spreads do not become more negative as retail participation increases. For example, in low-BTM firms, the MAX spread based on AbnRetail(P&S) is -0.39% for low-retail stocks and -0.42% for high-retail stocks, while corresponding patterns for retail flow measures show no monotonic increase. In medium- and high-BTM terciles, the same qualitative result holds, MAX underperformance is present but its magnitude is largely unrelated to the level of retail participation. The value-weighted results in Appendix Tables B10 and B11 mirror these findings, indicating that they are not driven by small or illiquid firms.

If the retail–MAX interaction were driven by classic lottery preferences concentrated in growth stocks, one would expect the MAX anomaly to be strongest precisely among low-BTM, high-retail portfolios. Instead, the data show that MAX underperformance is pervasive across both growth and value stocks and that retail participation does not systematically exacerbate this effect within

any valuation group. At the same time, retail participation continues to be associated with higher average returns within BTM terciles, consistent with earlier sections. This combination of results is difficult to reconcile with standard lottery-overpricing narratives, which predict that retail demand for extreme payoffs should inflate prices and reduce subsequent returns most strongly where lottery characteristics and retail participation overlap.

6. Conclusion

This paper examines the interaction between retail investor participation and lottery-like return characteristics in the cross-section of stock returns. Using multiple measures of retail participation, a comprehensive set of portfolio sorts, Fama–MacBeth regressions, earnings-announcement decompositions, and extensive cross-sectional analyses, we document three central findings. First, retail participation is associated with positive and economically meaningful future returns. Second, the MAX anomaly, stocks experiencing extreme recent daily returns subsequently underperform, remains robust and economically large. Third, and most importantly, these two effects are largely independent, retail participation does not amplify the MAX anomaly, nor does MAX explain the retail return premium.

Across all specifications, stocks with higher retail participation earn higher subsequent returns. This pattern holds in single sorts, double sorts with MAX, Fama–MacBeth regressions with extensive controls, earnings and non-earnings weeks, and triple sorts conditioning on firm size and book-to-market. The magnitude of the retail premium is economically meaningful, on the order of 40–65 basis points per week in high-minus-low portfolio spreads and remains statistically significant after adjusting for standard risk factors.

The results suggest that retail participation reflects more than uninformed speculation. Instead, retail trading appears to proxy for systematic demand pressure, information acquisition, or behavioral attention mechanisms that are not immediately arbitrated away. Importantly, the retail premium is present both during earnings-announcement weeks, when information arrival is concentrated, and during non-announcement weeks, indicating that retail investors are not merely reacting mechanically to public news. Rather, retail activity may capture dispersed private signals, expectation formation about near-term fundamentals, or gradual price discovery in segments of the market where institutional attention is limited.

The persistence of the retail premium after conditioning on size, valuation, momentum, volatility, and skewness further implies that retail participation captures a distinct dimension of return predictability. Retail investors may be early adopters of themes, products, or narratives that later diffuse more broadly, generating positive returns before prices fully adjust. In this sense, retail participation behaves less like noise trading and more like a slow-moving demand factor.

Classic lottery models predict a tight link between retail demand and lottery-like stocks. Under this view, retail investors disproportionately chase extreme upside payoffs, pushing prices above

fundamentals and generating subsequent negative returns. If this mechanism were dominant, the MAX anomaly should be strongest precisely among stocks with high retail participation.

Our evidence does not support this prediction. Across double and triple sorts, the MAX anomaly does not systematically intensify with retail participation. In many cases, the MAX penalty is largest among low-retail stocks and weekended, or unchanged, among high-retail stocks. This pattern holds across alternative retail measures, earnings and non-earnings weeks, and conditioning on size and book-to-market. Fama–MacBeth regressions confirm that the interaction between retail participation and MAX is statistically weak and economically modest once controls are included.

This lack of interaction implies that retail participation and lottery-like return realizations represent distinct channels of return predictability. Retail investors are not the marginal traders driving MAX mispricing, nor is the retail return premium simply compensation for bearing lottery risk. Instead, MAX appears to reflect episodic overreaction to extreme realizations, while retail participation reflects a broader, more persistent demand or information channel.

Our findings have several implications for asset pricing theory and models of retail behavior. First, they challenge the view that retail investors are primarily lottery-driven speculators whose trades systematically destroy value. Instead, retail participation is associated with positive abnormal returns, even after accounting for lottery-like characteristics. Second, the results imply that lottery preferences alone cannot explain the cross-section of returns, as lottery-like payoffs (MAX) are penalized but this penalty is largely independent of retail demand.

Our evidence supports a framework in which retail investors play a role in price formation. Retail trading may contribute to temporary mispricing in extreme-return stocks, but it also appears to facilitate information diffusion and demand discovery in a wide range of firms. Asset pricing models that treat retail investors as homogeneous noise traders may therefore miss an important dimension of market behavior.

Finally, the independence between retail participation and MAX suggests that future research should distinguish between persistent demand-based predictors and episodic return realizations. Failing to distinguish between the two risks misattributes the source of return predictability. Understanding when retail trading reflects informed demand versus speculative attention remains a promising avenue for future work.

7. References

- Allredge, D. M. (2020). Institutional trading, investor sentiment, and lottery-like stock preferences. *Financial Review*, 55(4), 603-624.
- Ang, A., Hodrick, R. J., Xing, Y., & Zhang, X. (2006). The cross-section of volatility and expected returns. *The Journal of Finance*, 61(1), 259-299.
- Ang, A., Hodrick, R. J., Xing, Y., & Zhang, X. (2009). High idiosyncratic volatility and low returns: International and further US evidence. *Journal of Financial Economics*, 91(1), 1-23.
- Bali, T. G., Hirshleifer, D., Peng, L., & Tang, Y. (2021). Attention, social interaction, and investor attraction to lottery stocks (No. w29543). *National Bureau of Economic Research*.
- Bali, T. G., Cakici, N., & Whitelaw, R. F. (2011). Maxing out: Stocks as lotteries and the cross-section of expected returns. *Journal of Financial Economics*, 99(2), 427-446.
- Barber, B. M., Huang, X., Odean, T., & Schwarz, C. (2022). Attention-induced trading and returns: Evidence from Robinhood users. *The Journal of Finance*, 77(6), 3141-3190.
- Barber, B. M., & Odean, T. (2008). All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *The Review of Financial Studies*, 21(2), 785-818.
- Barberis, N., & Huang, M. (2008). Stocks as lotteries: The implications of probability weighting for security prices. *American Economic Review*, 98(5), 2066-2100.
- Boyer, B., Mitton, T., & Vorkink, K. (2010). Expected idiosyncratic skewness. *The Review of Financial Studies*, 23(1), 169-202.
- Conrad, J., Kapadia, N., & Xing, Y. (2014). Death and jackpot: Why do individual investors hold overpriced stocks?. *Journal of Financial Economics*, 113(3), 455-475.
- Da, Z., Gurun, U. G., & Warachka, M. (2014). Frog in the pan: Continuous information and momentum. *The review of financial studies*, 27(7), 2171-2218.
- DellaVigna, S., & Pollet, J. M. (2009). Investor inattention and Friday earnings announcements. *The journal of finance*, 64(2), 709-749.
- Dimson, E. (1979). Risk measurement when shares are subject to infrequent trading. *Journal of Financial Economics*, 7(2), 197-226.
- Eaton, G. W., Green, T. C., Roseman, B., & Wu, Y. (2021). Zero-commission individual investors, high frequency traders, and stock market quality. *High Frequency Traders, and Stock Market Quality* (January 2021), 3.
- Fama, E. F., & MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3), 607-636.

- Fama, E. F., & French, K. R. (1992). The cross-section of expected stock returns. *The Journal of Finance*, 47(2), 427-465.
- Fama, E. F., & French, K. R. (1996). Multifactor explanations of asset pricing anomalies. *The Journal of Finance*, 51(1), 55-84.
- Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1-22.
- Fong, W. M., & Toh, B. (2014). Investor sentiment and the MAX effect. *Journal of Banking & Finance*, 46, 190-201.
- Goetzmann, W. N., & Kumar, A. (2008). Equity portfolio diversification. *Review of Finance*, 12(3), 433-463.
- Greenwood, R., & Nagel, S. (2009). Inexperienced investors and bubbles. *Journal of Financial Economics*, 93(2), 239-258.
- Harvey, C. R., & Siddique, A. (2000). Conditional skewness in asset pricing tests. *The Journal of Finance*, 55(3), 1263-1295.
- Hong, H., & Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *The Journal of finance*, 54(6), 2143-2184.
- Hou, K., Xue, C., & Zhang, L. (2015). Digesting anomalies: An investment approach. *The Review of Financial Studies*, 28(3), 650-705.
- Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *The Journal of Finance*, 45(3), 881-898.
- Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1), 65-91.
- Kumar, A. (2009). Who gambles in the stock market? *The Journal of Finance*, 64(4), 1889-1933.
- Lehmann, B. N. (1990). Fads, martingales, and market efficiency. *The Quarterly Journal of Economics*, 105(1), 1-28.
- Newey, W. K., and K. D. West. (1987). A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. *Econometrica*, 55(3), 703-708
- Nguyen, H. T., & Truong, C. (2018). When are extreme daily returns not lottery? At earnings announcements!. *Journal of Financial Markets*, 41, 92-116.
- Newey, W.K., West, K.D., 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Ozik, G., Sadka, R., & Shen, S. (2021). Flattening the illiquidity curve: Retail trading during the COVID-19 lockdown. *Journal of Financial and Quantitative Analysis*, 56(7), 2356-2388.

Seasholes, M. S., & Wu, G. (2007). Predictable behavior, profits, and attention. *Journal of Empirical Finance*, 14(5), 590-610.

Scholes, M., & Williams, J. (1977). Estimating betas from nonsynchronous data. *Journal of Financial Economics*, 5(3), 309-327

Sims, C. A. (2003). Implications of rational inattention. *Journal of monetary Economics*, 50(3), 665-690.

Tversky, A., & Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5, 297-323.

Welch, I. (2022). The wisdom of the Robinhood crowd. *The Journal of Finance*, 77(3), 1489-1527.

Table 1**Summary Statistics for Quintile Portfolios Sorted by Abnormal Retail Ownership (Size–Price Adjusted)**

This table reports summary statistics for quintile portfolios formed by sorting stocks on abnormal retail ownership adjusted for size and price (AbnRetail(P&S)). Portfolios are formed weekly from May 2018 to August 2020. In each week, stocks are sorted into five portfolios based on their AbnRetail(P&S) measured at the end of the previous week, with Portfolio 1 (Low) containing stocks with the lowest abnormal retail ownership and Portfolio 5 (High) containing stocks with the highest. For each quintile, the table reports the time-series average of the within-week median values of stock characteristics over the sample period. Reported variables include AbnRetail(P&S), maximum daily return (MAX), idiosyncratic volatility (IVOL), idiosyncratic skewness (ISKEW), the composite lottery measure, Z score, the natural logarithm of market capitalization (SIZE), market beta (BETA), book-to-market ratio (BTM), cumulative return over the 11 weeks prior to portfolio formation excluding the most recent week (MOM), the return in the portfolio formation week (REV), and share turnover (TURN). The final column reports the average number of stocks in each portfolio.

rank	AbnRetail(P&S)	MAX	IVOL	ISKEW	Zscore	BETA	Size	BTM	MOM	REV	TURN	#share
Low	-2.19	2.58	1.62	7.69	-0.50	0.74	5.99	0.97	-2.25	-0.10	5.74	726
2	-0.84	3.28	2.18	8.55	-0.12	1.00	6.65	0.80	-1.20	-0.06	6.95	726
3	-0.08	3.73	2.56	9.26	0.06	1.01	6.63	0.70	1.13	0.10	8.65	726
4	0.71	4.25	3.04	14.03	0.22	1.06	6.73	0.72	3.09	0.30	12.48	726
High	2.23	5.55	4.04	20.54	0.38	1.19	7.45	1.01	7.99	0.77	65.02	726

Table 2**Single-Sorted Portfolio Returns by MAX and Retail Participation Measures**

This table reports average weekly excess returns for quintile portfolios formed by sorting stocks on MAX and four distinct measures of retail participation. Portfolios are formed weekly from May 2018 to August 2020. In each week, stocks are sorted into five portfolios based on the lagged value of the sorting variable, with “Low” and “High” denoting the lowest and highest quintiles, respectively. Panel A reports equal-weighted portfolio returns, and Panel B reports value-weighted portfolio returns. For each sorting variable, the table reports average portfolio returns, the High–Low return spread, and corresponding t-statistics (in parentheses). Risk-adjusted performance is evaluated using alphas from the Fama–French six-factor model (FF6) and the q-factor model (QF), with Newey–West (1987) adjusted t-statistics reported below each estimate. The MAX column reports portfolios sorted on the maximum daily return within the previous week. The remaining columns report portfolios sorted on retail participation measures. NetRetail(SharesOut) is the weekly change in share-adjusted Robinhood ownership, $\Delta(Holders_{i,w}/SharesOut_{i,w})$. AbnRetail(P&S) is the residual from the weekly cross-sectional regression of $\log(Holders_{i,w})$ on $\log(MktValue_{i,w})$ and $\log(Price_{i,w})$. AbnRetailFlow(P&S) is the residual from the weekly cross-sectional regression of $\Delta(Holders_{i,w}/SharesOut_{i,w})$ on lagged $\log(MktValue)$ and $\log(Price)$. AbnRetailFlow(Micro) is the residual from the analogous regression additionally controlling for lagged $Volatility_{21d}$, Turnover, and $PastRet_{21d}$.

	MAX	NetRetail(Sharesout)	AbnRetail(P&S)	AbnRetailFlow(P&S)	AbnRetailFlow(Micro)
Panel A: Equal Weighted					
Low	0.49	0.28	0.06	0.13	0.11
2	0.32	0.17	0.14	0.17	0.13
3	0.25	0.19	0.23	0.23	0.18
4	0.22	0.28	0.39	0.28	0.26
High	0.14	0.59	0.55	0.62	0.54
High-Low	-0.35 (-0.87)	0.32 (2.12)	0.49 (2.65)	0.48 (2.10)	0.43 (2.21)
α_{FF6}	-0.27 (-0.98)	0.12 (0.88)	0.43 (2.17)	0.26 (1.68)	0.34 (2.00)
α_{QF}	-0.27 (-1.07)	0.32 (1.84)	0.46 (1.97)	0.39 (1.89)	0.24 (1.78)
Panel B: Value Weighted					
Low	0.42	0.20	0.08	0.12	0.12
2	0.31	0.19	0.15	0.21	0.26
3	0.29	0.23	0.17	0.27	0.27
4	0.23	0.27	0.22	0.35	0.35
High	0.08	0.37	0.39	0.46	0.41
High-Low	-0.34 (-0.96)	0.18 (1.48)	0.31 (2.66)	0.34 (1.28)	0.28 (1.61)
α_{FF6}	-0.26 (-0.94)	0.27 (0.93)	0.37 (2.28)	0.22 (0.91)	0.31 (1.48)
α_{QF}	-0.25 (-1.00)	0.24 (1.54)	0.30 (2.54)	0.26 (1.10)	0.15 (0.88)

Table 3**Single-Sorted Portfolio Returns Based on Abnormal Retail Ownership (Size–Price Adjusted)**

This table reports cumulative future returns for quintile portfolios formed by sorting stocks on abnormal retail ownership adjusted for size and price, AbnRetail(P&S). Portfolios are formed weekly from May 2018 to August 2020. In each week, stocks are sorted into five portfolios based on AbnRetail(P&S) measured at the end of the previous week, with Portfolio 1 (Low) containing stocks with the lowest abnormal retail ownership and Portfolio 5 (High) containing stocks with the highest. Panel A reports equal-weighted returns and Panel B reports value-weighted returns. Columns 1 through 11 report cumulative portfolio returns over holding horizons ranging from one to eleven weeks ahead. The High–Low row reports the return spread between the highest and lowest abnormal retail ownership portfolios, with Newey–West adjusted t -statistics reported below. The rows labeled α_{FF6} report alphas from time-series regressions on the Fama–French six-factor model, and α_{QF} reports alphas from regressions on the q-factor model. All returns and alphas are expressed in percentage terms.

	1	2	3	4	5	6	7	8	9	10	11
Panel A: Equal Weighted											
Low	0.06	0.05	0.03	-0.02	-0.07	-0.19	-0.29	-0.43	-0.48	-0.62	-0.66
2	0.14	0.22	0.30	0.32	0.38	0.36	0.36	0.30	0.33	0.29	0.34
3	0.23	0.38	0.54	0.68	0.84	0.94	1.02	1.03	1.14	1.19	1.33
4	0.39	0.75	1.09	1.41	1.75	2.00	2.30	2.46	2.70	2.88	3.05
High	0.55	1.04	1.51	1.98	2.42	2.75	3.06	3.41	3.86	4.14	4.47
High-Low	0.49 (2.65)	0.99 (2.72)	1.48 (2.73)	2.00 (2.73)	2.49 (2.72)	2.95 (2.71)	3.36 (2.67)	3.84 (2.65)	4.35 (2.64)	4.76 (2.63)	5.13 (2.61)
α_{FF6}	0.43 (2.17)	0.80 (1.56)	1.26 (1.64)	1.65 (1.70)	1.94 (1.66)	2.21 (1.57)	2.60 (1.56)	2.88 (1.55)	3.12 (1.52)	3.50 (1.54)	3.94 (1.53)
α_{QF}	0.46 (1.97)	0.96 (1.59)	1.42 (1.66)	1.95 (1.74)	2.41 (1.70)	2.75 (1.64)	3.23 (1.66)	3.74 (1.66)	4.14 (1.64)	4.50 (1.67)	4.67 (1.64)
Panel B: Value Weighted											
Low	0.08	0.12	0.17	0.21	0.22	0.16	0.11	0.02	0.00	-0.05	-0.02
2	0.15	0.27	0.41	0.54	0.67	0.75	0.82	0.84	0.91	0.95	1.05
3	0.17	0.20	0.31	0.39	0.47	0.49	0.52	0.51	0.55	0.54	0.61
4	0.22	0.41	0.60	0.76	0.95	1.10	1.21	1.30	1.41	1.49	1.64
High	0.39	0.78	1.14	1.51	1.88	2.24	2.56	2.84	3.15	3.43	3.75
High-Low	0.31 (2.66)	0.66 (2.76)	0.97 (2.88)	1.30 (2.97)	1.66 (3.06)	2.08 (3.19)	2.45 (3.24)	2.81 (3.26)	3.14 (3.30)	3.49 (3.35)	3.77 (3.33)
α_{FF6}	0.37 (2.28)	0.73 (2.62)	1.12 (2.74)	1.32 (2.78)	1.59 (2.90)	1.99 (3.02)	2.48 (3.14)	2.70 (3.17)	3.10 (3.28)	3.39 (3.23)	3.77 (3.22)
α_{QF}	0.30 (2.54)	0.72 (2.76)	1.04 (2.88)	1.35 (2.90)	1.69 (3.04)	1.98 (3.09)	2.43 (3.27)	2.80 (3.36)	3.11 (3.45)	3.48 (3.48)	3.71 (3.38)

Table 4**Double-Sorted Portfolios on Abnormal Retail Ownership and MAX**

This table reports average future returns for portfolios double-sorted on abnormal retail ownership (size–price adjusted) and MAX, the maximum daily return in the prior week. Portfolios are formed weekly from May 2018 to August 2020. At the beginning of each week, stocks are first sorted into five quintiles based on abnormal retail ownership, AbnRetail(P&S), measured at the end of the previous week. Within each retail-ownership quintile, stocks are then independently sorted into five quintiles based on MAX, computed as the highest daily return within the prior week. Panel A reports equal-weighted portfolio returns, and Panel B reports value-weighted portfolio returns. Each cell reports the average weekly excess return (in percent). The rows labeled E[re] report the return spread between the high-MAX and low-MAX portfolios within each retail-ownership quintile. Corresponding Fama–French six-factor alphas (α_{FF6}) and q-factor alphas (α_{QF}) are reported below, with Newey–West t-statistics (lag 4) in parentheses.

	Low MAX	2	3	4	High MAX	E[re]	α_{FF6}	α_{QF}
Panel (A): Equal Weighted								
Low AbnRetail(P&S)	0.20	0.13	0.16	0.02	-0.35	-0.55 (-1.53)	-0.44 (-1.61)	-0.51 (-1.8)
2	0.21	0.20	0.12	0.19	-0.05	-0.26 (-1.03)	-0.36 (-1.56)	-0.33 (-1.49)
3	0.63	0.26	0.20	0.04	0.00	-0.62 (-1.81)	-0.46 (-1.66)	-0.60 (-1.95)
4	0.70	0.32	0.29	0.21	0.36	-0.34 (-0.77)	-0.41 (-0.99)	-0.37 (-0.92)
High AbnRetail(P&S)	0.91	0.47	0.68	0.57	0.31	-0.60 (-1.82)	-0.65 (-1.89)	-0.63 (-1.86)
E[re]	0.71 (2.29)	0.34 (1.47)	0.52 (1.79)	0.55 (1.68)	0.66 (2.03)			
α_{FF6}	0.57 (2.25)	0.34 (1.35)	0.33 (1.51)	0.56 (1.83)	0.55 (2.10)			
α_{QF}	0.60 (2.33)	0.36 (1.41)	0.40 (1.46)	0.53 (1.66)	0.58 (1.98)			
Panel (B): Value Weighted								
Low AbnRetail(P&S)	0.05	0.04	0.17	0.11	-0.06	-0.11 (-0.45)	-0.24 (-0.85)	-0.20 (-0.77)
2	0.18	0.12	0.02	0.10	0.05	-0.13 (-0.66)	-0.33 (-1.33)	-0.28 (-1.2)
3	0.30	0.13	0.09	0.09	0.15	-0.16 (-0.59)	-0.08 (-0.31)	-0.19 (-0.88)
4	0.44	0.17	0.14	0.14	0.11	-0.33 (-1.31)	-0.31 (-1.15)	-0.36 (-1.42)
High AbnRetail(P&S)	0.62	0.48	0.36	0.18	0.50	-0.11 (-0.26)	-0.08 (-0.17)	0.01 (0.03)
E[re]	0.57 (2.54)	0.44 (2.47)	0.19 (1.04)	0.06 (0.28)	0.57 (2.38)			
α_{FF6}	0.55 (2.02)	0.49 (2.19)	0.15 (0.7)	0.03 (0.12)	0.71 (2.22)			
α_{QF}	0.50 (2.71)	0.40 (2.37)	0.10 (0.53)	0.02 (0.11)	0.70 (2.12)			

Table 5**MAX Return Spreads across Retail Participation Quintiles**

This table reports High-MAX minus Low-MAX return spreads within quintiles of retail participation. Stocks are first sorted weekly into quintiles based on a given retail participation measure measured at the end of the previous week. Within each retail quintile, stocks are then sorted into MAX quintiles based on the maximum daily return in the prior week. For each retail quintile, the table reports the difference in average future returns between the highest-MAX and lowest-MAX portfolios. Panel A presents equal-weighted portfolio results, and Panel B presents value-weighted results. For each retail measure, E[re] reports the raw High-MAX minus Low-MAX return spread, and the second column reports the corresponding alpha from the Fama–French six-factor model (α_{FF6}). Newey–West t-statistics (with appropriate lag length) are reported in parentheses. The retail participation measures include net retail buying scaled by shares outstanding (NetRetail(SharesOut)), abnormal retail ownership adjusted for size and price (AbnRetail(P&S)), abnormal retail flows adjusted for size and price (AbnRetailFlow(P&S)), and microstructure-adjusted abnormal retail flows (AbnRetailFlow(Micro)). Variable construction is described in Section 4.1.

	NetRetail(SharesOut)		AbnRetail(P&S)		AbnRetailFlow(P&S)		AbnRetailFlow(Micro)	
	E[re]	α_{FF6}	E[re]	α_{FF6}	E[re]	α_{FF6}	E[re]	α_{FF6}
Panel (A): Equal Weighted								
Low	-0.17 (-0.77)	-0.22 (-1.12)	-0.55 (-1.53)	-0.44 (-1.61)	-0.65 (-1.99)	-0.66 (-2.54)	-0.52 (-1.23)	-0.54 (-1.63)
2	-0.41 (-1.42)	-0.40 (-1.38)	-0.26 (-1.03)	-0.36 (-1.56)	-0.36 (-1.01)	-0.33 (-1.17)	-0.37 (-1.02)	-0.28 (-0.94)
3	-0.28 (-1.08)	-0.30 (-1.39)	-0.62 (-1.81)	-0.46 (-1.66)	-0.31 (-1.03)	-0.29 (-1)	-0.31 (-1.2)	-0.31 (-1.33)
4	-0.35 (-0.91)	-0.24 (-0.75)	-0.34 (-0.77)	-0.41 (-0.99)	-0.27 (-1.02)	-0.29 (-1.12)	-0.29 (-1.16)	-0.31 (-1.24)
High	-0.90 (-1.89)	-0.83 (-2.20)	-0.60 (-1.82)	-0.65 (-1.89)	-0.57 (-1.42)	-0.53 (-1.56)	-0.57 (-1.42)	-0.51 (-1.56)
Panel (B): Value Weighted								
Low	-0.35 (-0.88)	-0.59 (-1.57)	-0.11 (-0.45)	-0.24 (-0.85)	-0.39 (-1.13)	-0.44 (-1.26)	-0.6 (-1.32)	-0.57 (-1.29)
2	-0.25 (-0.97)	-0.22 (-0.80)	-0.13 (-0.66)	-0.33 (-1.33)	-0.7 (-1.8)	-0.71 (-2.06)	-0.16 (-0.57)	-0.27 (-0.87)
3	-0.14 (-0.67)	-0.17 (-0.83)	-0.16 (-0.59)	-0.08 (-0.31)	-0.28 (-0.91)	-0.35 (-1.21)	-0.04 (-0.15)	0.03 (0.14)
4	-0.15 (-0.50)	-0.20 (-0.69)	-0.33 (-1.31)	-0.31 (-1.15)	-0.21 (-0.77)	-0.14 (-0.49)	-0.08 (-0.4)	-0.07 (-0.28)
High	-0.40 (-1.27)	-0.41 (-1.21)	-0.11 (-0.26)	-0.08 (-0.17)	-0.33 (-1.03)	-0.25 (-0.82)	-0.36 (-1.22)	-0.36 (-1.17)

Table 6**Fama–MacBeth Cross-Sectional Regressions: Retail Participation, MAX, and Future Returns**

This table reports results from weekly Fama–MacBeth (1973) cross-sectional regressions of one-week-ahead stock excess returns on abnormal retail ownership (size–price adjusted), MAX, and their interaction. In each week from May 2018 to August 2020, stock-level excess returns in week $t+1$ are regressed on variables measured at the end of week t . The main specification includes AbnRetail(P&S), defined as the residual from a cross-sectional regression of the log number of Robinhood holders on log market capitalization and log price, MAX, defined as the maximum daily return during the previous week, and their interaction $\text{AbnRetail(P\&S)} \times \text{MAX}$. Subsequent columns incrementally add standard firm-level controls: market beta (BETA), size (SIZE), book-to-market (BTM), cumulative return over weeks $t-1$ to $t-2$ (MOM), the return in week t (REV), share turnover (TURN), idiosyncratic volatility (IVOL), and idiosyncratic skewness (ISKEW). Reported coefficients are time-series averages of the weekly slope estimates. t -statistics, shown in parentheses, are computed from the time series of weekly coefficients using Newey–West standard errors to account for serial correlation. All coefficients are expressed in percent per week.

AbnRetail(P&S) $\times$ MAX	AbnRetail(P&S)	MAX	BETA	SIZE	BTM	MOM	REV	TURN	IVOL	ISKEW
2.49 (0.76)	1.50 (2.51)	-3.12 (-4.86)								
2.65 (0.78)	1.52 (2.55)	-3.45 (-4.17)	0.14 (0.43)	-0.48 (-0.92)						
2.13 (0.64)	1.53 (2.56)	-3.43 (-4.26)	0.12 (0.37)	-0.44 (-0.90)	1.25 (3.07)					
2.63 (0.76)	1.63 (2.57)	-3.64 (-4.26)	0.33 (1.28)	-0.37 (-0.81)	1.28 (3.14)	4.76 (1.72)				
2.99 (0.86)	1.61 (2.57)	-3.62 (-4.27)	0.35 (1.29)	-0.35 (-0.77)	1.28 (3.14)	4.72 (1.99)	-1.39 (-3.29)			
4.55 (1.52)	1.58 (2.62)	-3.56 (-4.20)	0.34 (1.30)	-0.34 (-0.73)	1.29 (3.14)	4.80 (2.01)	-0.83 (-2.16)	0.02 (1.94)		
4.61 (1.44)	1.53 (2.63)	-3.70 (-4.59)	0.31 (1.41)	-0.30 (-0.61)	1.29 (3.17)	4.41 (1.89)	-0.92 (-2.26)	0.02 (2.03)	-2.15 (-2.33)	
4.73 (1.50)	1.54 (2.63)	-3.76 (-4.65)	0.30 (1.37)	-0.30 (-0.60)	1.29 (3.17)	4.64 (1.97)	-1.47 (-3.24)	0.02 (2.03)	-2.14 (-2.33)	2.22 (1.69)

Table 7**Retail Participation and Future Returns during Earnings-Announcement and Non-Earnings Weeks**

This table reports single-sorted portfolio returns based on retail participation measures, separately for earnings-announcement (EA) weeks and non-earnings-announcement (NOEA) weeks. Portfolios are formed weekly from May 2018 to August 2020 by sorting stocks into quintiles based on each retail participation measure measured at the end of the previous week. The four retail measures are NetRetail(SharesOut), AbnRetail(P&S), AbnRetailFlow(P&S), and AbnRetailFlow(Micro), as defined in Section 4.1. Panel A reports results for EA weeks, defined as weeks in which firms announce quarterly earnings. Panel B reports results for NOEA weeks, defined as all remaining weeks. For each retail quintile, the table reports average one-week-ahead excess returns. The row labeled E[re] reports the high-minus-low return spread across retail participation quintiles. The rows labeled α_{FF6} and α_{QF} report alphas from the Fama–French six-factor model and the q-factor model, respectively. All portfolios are equal-weighted. t-statistics, shown in parentheses, are computed using Newey–West standard errors to account for serial correlation. Returns and alphas are expressed in percent per week.

	NetRetail(SharesOut)	AbnRetail(P&S)	AbnRetailFlow(P&S)	AbnRetailFlow(Micro)
Panel (A):				
Equal Weighted Earning Announcement Window				
Low	-0.08	-0.13	0.04	0.00
2	0.05	0.13	0.19	0.16
3	0.31	0.30	0.26	0.17
4	0.41	0.36	0.32	0.23
High	0.45	0.41	0.44	0.41
E[re]	0.53 (2.32)	0.54 (2.38)	0.40 (2.11)	0.41 (2.16)
α_{FF6}	0.49 (2.08)	0.59 (2.60)	0.35 (1.62)	0.43 (1.93)
α_{QF}	0.55 (1.83)	0.54 (2.31)	0.42 (1.86)	0.46 (1.96)
Panel (B): Equal Weighted NO-Earning Announcement Window				
Low	0.14	0.01	-0.05	-0.05
2	0.04	0.03	0.02	0.05
3	0.09	0.13	0.16	0.11
4	0.13	0.18	0.28	0.25
High	0.53	0.58	0.58	0.59
E[re]	0.39 (2.03)	0.57 (2.12)	0.63 (2.48)	0.64 (2.75)
α_{FF6}	0.34 (1.40)	0.33 (2.06)	0.46 (2.34)	0.45 (2.58)
α_{QF}	0.29 (1.55)	0.44 (1.99)	0.62 (2.27)	0.61 (2.25)

Table 8**Retail Participation and the MAX Effect during Earnings-Announcement and Non-Earnings Weeks**

This table reports double-sorted portfolio tests examining the interaction between retail participation and the MAX anomaly, separately for earnings-announcement (EA) weeks and non-earnings-announcement (NOEA) weeks. Portfolios are formed weekly from May 2018 to August 2020. In each week, stocks are first sorted into quintiles based on a retail participation measure measured at the end of the previous week, NetRetail(SharesOut), AbnRetail(P&S), AbnRetailFlow(P&S), or AbnRetailFlow(Micro). Within each retail quintile, stocks are then sorted into quintiles based on MAX, defined as the maximum daily return during the previous week. For each retail participation quintile, the table reports equal weighted high-minus-low MAX return spread (High MAX – Low MAX). The column labeled E[re] reports the average one-week-ahead excess return spread, while α_{FF6} reports the corresponding alpha from the Fama–French six-factor model. Panel A reports result for EA weeks, defined as weeks containing a firm's quarterly earnings announcement. Panel B reports result for NOEA weeks. t-statistics, shown in parentheses, are computed using Newey–West standard errors to account for serial correlation. All returns and alphas are expressed in percent per week.

	NetRetail(SharesOut)		AbnRetail(P&S)		AbnRetailFlow(P&S)		AbnRetailFlow(Micro)	
	E[re]	α_{FF6}	E[re]	α_{FF6}	E[re]	α_{FF6}	E[re]	α_{FF6}
Panel (A): Equal Weighted Earning Announcement Window								
Low	-0.28 (-1.80)	-0.56 (-2.08)	-0.20 (-2.33)	-0.95 (-2.59)	-0.25 (-0.95)	-0.33 (-0.94)	-0.23 (-0.61)	-0.49 (-0.90)
2	-0.24 (-0.69)	-0.35 (-0.96)	-0.07 (-0.44)	-0.29 (-0.89)	-0.36 (-1.29)	-0.50 (-1.31)	-0.33 (-1.67)	-0.82 (-1.91)
3	-0.29 (-0.57)	-0.33 (-0.59)	-0.06 (-1.12)	-0.56 (-1.63)	-0.16 (-0.90)	-0.42 (-1.12)	0.15 (0.56)	-0.01 (-0.02)
4	-0.10 (-0.21)	-0.31 (-0.80)	-0.09 (-0.58)	-0.36 (-0.98)	-0.37 (-2.15)	-0.74 (-2.45)	-0.07 (-0.92)	-0.19 (-1.40)
High	-0.27 (-1.20)	-0.43 (-0.82)	-0.10 (-1.19)	-0.28 (-0.59)	0.05 (0.10)	-0.12 (-0.23)	-0.06 (-0.12)	0.12 (0.20)
Panel (B): Equal Weighted NO-Earning Announcement Window								
Low	-0.46 (-1.95)	-0.44 (-1.96)	-0.43 (-1.71)	-0.44 (-1.77)	-0.63 (-2.11)	-0.63 (-2.57)	-0.53 (-1.47)	-0.68 (-2.01)
2	-0.38 (-1.32)	-0.31 (-1.45)	-0.46 (-1.67)	-0.43 (-1.93)	-0.28 (-0.84)	-0.23 (-0.75)	-0.39 (-1.64)	-0.36 (-1.68)
3	-0.45 (-1.92)	-0.45 (-1.89)	-0.41 (-1.68)	-0.39 (-1.69)	-0.35 (-1.48)	-0.39 (-1.59)	-0.14 (-0.57)	-0.01 (-0.06)
4	-0.54 (-1.59)	-0.48 (-1.71)	-0.42 (-1.09)	-0.37 (-1.01)	-0.13 (-0.51)	-0.14 (-0.53)	-0.25 (-1.20)	-0.20 (-0.92)
High	-0.70 (-1.39)	-0.59 (-1.45)	-0.57 (-1.12)	-0.52 (-1.25)	-0.65 (-1.39)	-0.63 (-1.67)	-0.54 (-1.22)	-0.49 (-1.41)

Table 9

Equal-weighted Triple-Sorted Portfolios by Size, Abnormal Retail Ownership, and MAX

This table reports equal-weighted triple-sorted portfolio results based on firm size, abnormal retail ownership (size–price adjusted), and the MAX characteristic, using weekly data from May 2018 to August 2020. In each week, stocks are first sorted into size terciles (Small, Medium, Big) based on market capitalization measured at the end of the previous week. Within each size tercile, stocks are then sorted into terciles based on abnormal retail ownership (AbnRetail(P&S)), measured at the end of the previous week. Finally, within each size–retail cell, stocks are further sorted into quintiles based on MAX, defined as the maximum daily return during the previous week. Portfolios are held for one week. Panel A, Panel B, and Panel C report results for small, medium, and big firms, respectively. Within each panel, the table reports average weekly excess returns for the resulting portfolios. The rows labeled E[re] report the high-minus-low MAX return spread (High MAX – Low MAX) within each retail ownership category. The rows labeled α_{FF6} and α_{QF} report the corresponding alphas from the Fama–French six-factor model and the Q-factor model, respectively. t-statistics, shown in parentheses, are computed using Newey–West standard errors. All returns and alphas are expressed in percent per week.

PANEL (A): Small Size								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	0.28	0.18	0.08	-0.19	-0.38	-0.66 (-2.26)	-0.96 (-2.34)	-1.09 (-2.3)
2	0.69	0.43	0.38	0.46	0.16	-0.53 (-1.88)	-0.4 (-1.54)	-0.49 (-1.45)
High AbnRetail(P&S)	1.06	0.89	0.81	0.76	0.42	-0.66 (-2.15)	-1.15 (-2.17)	-1.22 (-2.34)
E[re]	0.78 (2.14)	0.71 (2.28)	0.73 (2.39)	0.95 (2.93)	0.80 (2.04)			
α_{FF6}	0.71 (2.24)	0.78 (2.03)	0.79 (2.07)	1.09 (2.86)	0.66 (2.17)			
α_{QF}	0.93 (2.27)	0.84 (2.12)	0.90 (2.22)	1.12 (2.53)	0.77 (1.97)			
PANEL (B): Medium Size								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	-0.07	-0.05	-0.05	-0.19	-0.45	-0.38 (-1.09)	-0.41 (-1.22)	-0.34 (-1.23)
2	0.08	0.14	0.07	-0.05	-0.42	-0.5 (-1.61)	-0.51 (-1.88)	-0.56 (-2.15)
High AbnRetail(P&S)	0.5	0.45	0.34	0.26	0.32	-0.18 (-0.44)	-0.04 (-0.1)	-0.15 (-0.39)
E[re]	0.57 (1.48)	0.51 (1.63)	0.39 (1.06)	0.45 (1.86)	0.77 (2.38)			
α_{FF6}	0.28 (1.43)	0.31 (1.64)	0.19 (0.71)	0.38 (2.1)	0.65 (2.11)			
α_{QF}	0.61 (1.56)	0.51 (1.58)	0.32 (0.94)	0.45 (1.89)	0.81 (2.11)			
PANEL (C): Big Size								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	0.05	-0.02	-0.01	0.01	-0.09	-0.14 (-0.62)	-0.11 (-0.53)	-0.16 (-0.84)
2	0.35	0.21	0.09	0.10	-0.09	-0.43 (-1.82)	-0.46 (-2.16)	-0.52 (-2.15)
High AbnRetail(P&S)	0.53	0.39	0.18	0.26	0.31	-0.22 (-0.64)	-0.06 (-0.2)	-0.2 (-0.65)
E[re]	0.48 (2.06)	0.41 (2.01)	0.19 (1.08)	0.25 (1.38)	0.41 (2.17)			
α_{FF6}	0.38 (2.17)	0.42 (2.02)	0.18 (1.13)	0.28 (1.79)	0.43 (2.15)			
α_{QF}	0.44 (2.07)	0.45 (2.32)	0.21 (1.18)	0.3 (1.53)	0.4 (2.10)			

Table 10**Equal-weighted MAX Return Spreads Conditional on Firm Size and Retail Participation**

This table reports equal-weighted high-minus-low MAX return spreads (High MAX – Low MAX) conditional on firm size and retail participation, following the same triple-sorting procedure as Table 9. Using weekly data from May 2018 to August 2020, stocks are first sorted each week into terciles based on market capitalization (Small, Medium, and Big), measured at the end of the previous week. Within each size tercile, stocks are then sorted into quintiles based on retail participation, measured at the end of the previous week using four alternative proxies: NetRetail(SharesOut), AbnRetail(P&S), AbnRetailFlow(P&S), and AbnRetailFlow(Micro). For expositional clarity, the table reports results for three representative retail participation quintiles (Low, 3, and High) within each size group. Within each size–retail cell, stocks are further sorted into quintiles based on MAX, defined as the maximum daily return during the previous week, and held for one week. The table reports the difference in average weekly excess returns between the highest-MAX and lowest-MAX portfolios ($E[re]$) as well as the corresponding Fama–French six-factor alphas (α_{FF6}). Panels A, B, and C present results for small, medium, and big firms, respectively. t-statistics, shown in parentheses, are computed using Newey–West standard errors. All returns and alphas are expressed in percent per week.

	NetRetail(SharesOut)		AbnRetail(P&S)		AbnRetailFlow(P&S)		AbnRetailFlow(Micro)	
	$E[re]$	α_{FF6}	$E[re]$	α_{FF6}	$E[re]$	α_{FF6}	$E[re]$	α_{FF6}
PANEL (A): Small Size								
Low	-0.55 (-0.96)	-0.42 (-0.89)	-0.66 (-2.26)	-0.96 (-2.34)	-1.08 (-2.59)	-0.95 (-3.15)	-0.83 (-1.73)	-0.81 (-2.03)
2	-0.33 (-1.21)	-0.27 (-1.07)	-0.53 (-1.58)	-0.40 (-1.54)	-0.45 (-1.58)	-0.47 (-1.52)	-0.70 (-2.54)	-0.62 (-2.52)
High	-1.12 (-2.08)	-1.03 (-1.85)	-0.66 (-2.15)	-1.15 (-2.17)	-1.16 (-1.62)	-1.13 (-1.50)	-0.76 (-1.29)	-0.62 (-1.2)
PANEL (B): Medium Size								
Low	-0.31 (-0.67)	-0.25 (-0.59)	-0.38 (-1.09)	-0.41 (-1.22)	-0.59 (-1.35)	-0.49 (-1.24)	-0.48 (-0.95)	-0.44 (-0.95)
2	-0.10 (-0.48)	-0.09 (-0.43)	-0.5 (-1.61)	-0.51 (-1.88)	-0.31 (-0.98)	-0.37 (-1.18)	-0.11 (-0.35)	-0.05 (-0.19)
High	-0.62 (-1.82)	-0.52 (-1.65)	-0.18 (-0.44)	-0.04 (-0.10)	-0.69 (-1.56)	-0.61 (-1.57)	-0.54 (-1.42)	-0.49 (-1.17)
PANEL (C): Big Size								
Low	-0.28 (-0.76)	-0.28 (-0.82)	-0.14 (-0.62)	-0.11 (-0.53)	-0.23 (-1.01)	-0.26 (-1.11)	-0.21 (-0.94)	-0.22 (-1.05)
2	-0.14 (-0.85)	-0.12 (-0.7)	-0.43 (-1.82)	-0.46 (-2.16)	-0.24 (-1.12)	-0.17 (-0.84)	-0.22 (-1.42)	-0.19 (-1.22)
High	-0.30 (-0.97)	-0.17 (-0.64)	-0.22 (-0.64)	-0.06 (-0.2)	-0.25 (-0.88)	-0.14 (-0.61)	-0.02 (-0.07)	0.05 (0.19)

Table 11

Equal-weighted Triple-Sorted Portfolios by Book-to-Market, Abnormal Retail Ownership, and MAX

This table reports equal-weighted triple-sorted portfolio results based on book-to-market (BTM), abnormal retail ownership (size–price adjusted), and the MAX characteristic, following the same procedure as Table 9 but replacing firm size with book-to-market. Using weekly data from May 2018 to August 2020, stocks are first sorted each week into terciles based on their book-to-market ratio, measured at the end of the previous week, corresponding to Low, Medium, and High BTM firms. Within each BTM tercile, stocks are then sorted into quintiles based on abnormal retail ownership (AbnRetail(P&S)), measured at the end of the previous week. Finally, within each BTM–retail cell, stocks are further sorted into quintiles based on MAX, defined as the maximum daily return during the previous week. All portfolios are held for one week. Panel A, Panel B, and Panel C report results for Low, Medium, and High book-to-market firms, respectively. Within each panel, the table reports average weekly excess returns for the resulting portfolios. The rows labeled E[re] report the high-minus-low MAX return spread (High MAX – Low MAX) within each retail ownership category. The rows labeled α_{FF6} and α_{QF} report the corresponding alphas from the Fama–French six-factor model and the Q-factor model, respectively. t-statistics, shown in parentheses, are computed using Newey–West standard errors. All returns and alphas are expressed in percent per week.

PANEL (A): Low Book to Market								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	0.42	0.29	0.26	0.18	0.03	-0.39 (-1.11)	-0.15 (-0.81)	-0.2 (-1.07)
2	0.51	0.42	0.14	0.12	0.12	-0.39 (-0.93)	-0.29 (-0.77)	-0.34 (-0.9)
High AbnRetail(P&S)	0.69	0.44	0.40	0.34	0.27	-0.42 (-0.34)	-0.19 (-0.6)	-0.1 (-0.3)
E[re]	0.27 (1.93)	0.16 (0.45)	0.14 (0.35)	0.16 (1.03)	0.24 (2.02)			
α_{FF6}	0.33 (2.23)	0.04 (0.19)	0.09 (0.42)	0.12 (0.72)	0.43 (2.18)			
α_{QF}	0.42 (1.94)	0.05 (0.21)	0.06 (0.29)	0.17 (0.87)	0.53 (2.15)			
PANEL (B): Medium Book to Market								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	0.04	0.06	-0.06	-0.07	-0.58	-0.63 (-2.72)	-0.64 (-3.02)	-0.67 (-3.11)
2	0.15	0.14	0.06	0.03	-0.04	-0.19 (-0.34)	-0.08 (-0.28)	-0.21 (-0.81)
High AbnRetail(P&S)	0.51	0.47	0.33	0.22	-0.06	-0.57 (-1.29)	-0.53 (-1.31)	-0.57 (-1.48)
E[re]	0.46 (1.74)	0.41 (1.68)	0.39 (1.47)	0.29 (0.95)	0.52 (1.35)			
α_{FF6}	0.33 (1.5)	0.57 (1.52)	0.44 (1.22)	0.22 (0.96)	0.44 (1.3)			
α_{QF}	0.41 (1.59)	0.61 (1.87)	0.46 (1.54)	0.33 (1.09)	0.52 (1.32)			
PANEL (C): High Book to Market								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	0.06	0.08	0.04	-0.18	-0.61	-0.68 (-2.52)	-0.71 (-2.69)	-0.68 (-2.76)
2	0.29	0.19	0.07	0.09	-0.29	-0.59 (-1.93)	-0.61 (-2.22)	-0.63 (-2.07)
High AbnRetail(P&S)	1.11	0.78	0.36	1.01	0.41	-0.70 (-1.48)	-0.45 (-0.94)	-0.56 (-1.12)
E[re]	1.05 (1.73)	0.7 (1.28)	0.33 (0.69)	1.19 (1.69)	1.02 (1.85)			
α_{FF6}	0.63 (1.57)	0.51 (1.21)	0.12 (0.34)	1.23 (1.49)	0.88 (1.82)			
α_{QF}	0.88 (1.68)	0.60 (1.25)	0.24 (0.58)	1.21 (1.5)	1.01 (1.93)			

Table 12

Equal-Weighted MAX Return Spreads Conditional on Book-to-Market and Retail Participation

This table reports equal-weighted high-minus-low MAX return spreads (High MAX – Low MAX) conditional on firm book-to-market ratios and retail participation. The table follows the same empirical design as Table 10, but replaces firm size with book-to-market as the conditioning variable. Using weekly data from May 2018 to August 2020, stocks are first sorted each week into terciles based on their book-to-market ratio (Low, Medium, and High). Within each book-to-market tercile, stocks are then sorted into quintiles based on retail participation, measured at the end of the previous week using four alternative proxies: NetRetail(SharesOut), AbnRetail(P&S), AbnRetailFlow(P&S), and AbnRetailFlow(Micro). For expositional clarity, the table reports results for three representative retail participation quintiles (Low, 3, and High) within each book-to-market group. Within each book-to-market–retail cell, stocks are further sorted into quintiles based on MAX, defined as the maximum daily return during the previous week, and held for one week. The table reports the difference in average weekly excess returns between the highest-MAX and lowest-MAX portfolios ($E[re]$) and the corresponding Fama–French six-factor alphas (α_{FF6}). Panels A, B, and C present results for Low, Medium, and High book-to-market stocks, respectively. t-statistics, shown in parentheses, are computed using Newey–West standard errors. All returns and alphas are expressed in percent per week.

	NetRetail(SharesOut)		AbnRetail(P&S)		AbnRetailFlow(P&S)		AbnRetailFlow(Micro)	
	$E[re]$	α_{FF6}	$E[re]$	α_{FF6}	$E[re]$	α_{FF6}	$E[re]$	α_{FF6}
PANEL (A): Low Book to Market								
Low	-0.49 (-1.04)	-0.41 (-0.89)	-0.39 (-1.11)	-0.15 (-0.81)	-0.54 (-1.72)	-0.51 (-1.99)	-0.62 (-1.14)	-0.6 (-1.24)
2	-0.20 (-0.68)	-0.23 (-0.88)	-0.39 (-0.93)	-0.29 (-0.77)	-0.09 (-0.32)	-0.02 (-0.11)	-0.37 (-1.22)	-0.38 (-1.33)
High	-0.45 (-0.94)	-0.32 (-0.78)	-0.42 (-0.34)	-0.19 (-0.6)	-0.47 (-1.32)	-0.46 (-1.45)	-0.49 (-1.28)	-0.33 (-1.03)
PANEL (B): Medium Book to Market								
Low	-0.72 (-1.94)	-0.61 (-2.02)	-0.63 (-2.72)	-0.64 (-3.02)	-0.42 (-1.23)	-0.44 (-1.62)	-0.49 (-1.31)	-0.48 (-1.55)
2	-0.3 (-2.13)	-0.28 (-2.14)	-0.10 (-0.34)	-0.08 (-0.28)	-0.27 (-1.11)	-0.25 (-1.03)	-0.12 (-0.51)	-0.12 (-0.63)
High	-0.25 (-0.62)	-0.33 (-0.89)	-0.56 (-1.29)	-0.53 (-1.31)	-0.26 (-1.17)	-0.39 (-1.68)	-0.68 (-2.59)	-0.69 (-2.98)
PANEL (C): High Book to Market								
Low	-0.61 (-0.96)	-0.60 (-1.13)	-0.68 (-2.52)	-0.71 (-2.69)	-0.67 (-1.82)	-0.54 (-2.12)	-0.55 (-1.18)	-0.57 (-1.47)
2	-0.57 (-2.15)	-0.53 (-2.12)	-0.59 (-1.93)	-0.61 (-2.22)	-0.47 (-1.66)	-0.58 (-1.74)	-0.4 (-2.39)	-0.46 (-2.52)
High	-0.83 (-1.82)	-0.69 (-1.43)	-0.70 (-1.48)	-0.45 (-0.94)	-0.72 (-1.67)	-0.55 (-1.24)	-0.29 (-0.77)	-0.11 (-0.35)

Appendix A

1- Retail Net Buying (Share-Adjusted).

Retail Net Buying is measured as the weekly change in share-adjusted retail ownership. Let $Holders_{i,t}$ denote the number of distinct Robinhood user accounts holding stock i at the end of week w , and let $SharesOut_{i,w}$ denote shares outstanding. The measure is defined as:

$$NetRetail(Sharesout)_{i,w} = \frac{Holders_{i,w}}{SharesOut_{i,w}} - \frac{Holders_{i,w-1}}{SharesOut_{i,w-1}}.$$

2- Abnormal Retail Ownership (Size-Price Adjusted).

To isolate retail ownership that exceeds what is mechanically implied by firm size and share price, we construct an abnormal retail ownership measure. For each trading day t , we estimate the following cross-sectional regression:

$$\log(Holders_{i,t}) = \alpha_t + \beta_1 \log(MktValue_{i,t}) + \beta_2 \log(Price_{i,t}) + \varepsilon_{i,t},$$

where $Holders_{i,t}$ is the number of Robinhood users holding stock i , $MktValue_{i,t}$ is the market capitalization, and $Price_{i,t}$ is the stock's closing price. Abnormal retail ownership is defined as the weekly average of regression residual:

$$AbnRetail_{i,t} = \frac{1}{D(w)} \sum_{t \in D(w)} \hat{\varepsilon}_{i,t}.$$

3- Abnormal Retail Flow (Size-Price Adjusted).

Abnormal Retail Flow isolates unexpected weekly changes in share-adjusted retail ownership after controlling for firm size and price. For each week w , we estimate:

$$\Delta \left(\frac{Holders_{i,w}}{SharesOut_{i,w}} \right) = \gamma_w + \delta_1 \log(MktValue_{i,w-1}) + \delta_2 \log(Price_{i,w-1}) + u_{i,w}.$$

The residual $\hat{u}_{i,w}$ is the abnormal retail flow.

4- Abnormal Retail Flow (Microstructure-Adjusted)

The microstructure-adjusted abnormal retail flow further removes predictable variation related to short-horizon trading conditions. Weekly changes in share-adjusted retail ownership are regressed on lagged firm characteristics:

$$\Delta \left(\frac{Holders_{i,w}}{SharesOut_{i,w}} \right) = \eta_w + \theta' [\log(MktValue), \log(Price), Volatility_{21d}, Turnover, PastRet_{21d}]_{i,w-1} + \varepsilon_{i,w}.$$

The residual $\hat{\varepsilon}_{i,w}$ defines the microstructure-adjusted abnormal retail flow.

5- Maximum Daily Return (MAX).

Bali, Cakici, and Whitelaw (2011) document a significant negative relation between a stock's maximum daily return over the previous month and its subsequent return. Following their approach, we use MAX as our primary proxy for lottery-like characteristics but adapt the definition to a weekly return horizon consistent with our empirical design.

Specifically, for each stock i and week t , MAX is defined as the maximum daily return realized within that week:

$$MAX_{i,t} = \max(R_{i,d}), d \in \text{week } t,$$

where $R_{i,d}$ denotes the daily return of stock i on trading day d , and the maximization is taken over all trading days in week t (typically five).

6- Market Beta (BETA)

To account for nonsynchronous trading between individual stocks and the market index, we estimate market beta using the Dimson (1979) and Scholes and Williams (1977) correction, which incorporates lagged, contemporaneous, and leading market returns. Specifically, for each stock i , we estimate the following regression using daily returns:

$$R_{i,d} - r_{f,d} = \alpha_i + \beta_{1,i}(R_{m,d-1} - r_{f,d-1}) + \beta_{2,i}(R_{m,d} - r_{f,d}) + \beta_{3,i}(R_{m,d+1} - r_{f,d+1}) + \varepsilon_{i,d},$$

where $R_{i,d}$ is the return on stock i on day d , $R_{m,d}$ is the market return on day d , and $r_{f,d}$ is the risk-free rate. The market return is proxied by the CRSP daily value-weighted index, and the risk-free rate is the one-month Treasury bill return obtained from Kenneth French's data library.

Unlike the original monthly implementation, we estimate this regression using daily returns over a rolling 22-trading-day window ending on the final trading day of the week t .

The market beta for the stock i in a week t is then defined as the sum of the three estimated coefficients:

$$\text{BETA}_{i,t} = \hat{\beta}_{1,i,t} + \hat{\beta}_{2,i,t} + \hat{\beta}_{3,i,t}.$$

7- Firm Size (SIZE).

Consistent with the existing asset-pricing literature, firm size is measured as the natural logarithm of the market value of equity. Market value is computed as the stock price multiplied by shares outstanding, expressed in millions of dollars.

To align the size measure with the weekly return horizon used in the empirical analysis, firm size for stock i in week t is calculated using information available at the end of the previous week ($t - 1$):

$$\text{SIZE}_{i,t} = \log (\text{Price}_{i,t-1} \times \text{SharesOut}_{i,t-1}).$$

8- Book-to-Market Ratio (BTM).

Following Fama and French (1992), the book-to-market ratio is constructed using accounting information from Compustat and market values from CRSP. For each firm, the book value of equity is defined as the book value of common equity plus balance-sheet deferred taxes, measured at the firm's most recent fiscal year-end.

Market equity is measured as the market value of equity at the end of December of the prior calendar year, consistent with the standard Fama–French timing convention. The book-to-market ratio for stock i in week t is therefore defined as:

$$\text{BTM}_{i,t} = \frac{\text{Book Equity}_{i,\text{FY}(t-1)}}{\text{Market Equity}_{i,\text{Dec}(t-1)}}.$$

9- Short-Term Reversal (REV)

For each stock i in formation week t , the reversal variable is defined as the stock's return in the most recent week (the week immediately preceding portfolio formation): $\text{REV}_{i,t} = r_{i,t-1}$, where $r_{i,t-1}$ is the simple return from week $t - 1$ (e.g., Monday–Friday) for stock i .

10- Momentum (MOM)

For each stock i in formation week t , momentum is defined as the cumulative return over the 11 weeks ending two weeks prior, i.e., we skip the most recent week to reduce microstructure/reversal effects. Formally,

$$\text{MOM}_{i,t} = \prod_{k=2}^{12} (1 + r_{i,t-k}) - 1,$$

where $r_{i,t-k}$ is the stock's simple return in week $t - k$.

11- Turnover.

Trading activity is measured using stock turnover, defined as trading volume scaled by shares outstanding. For each stock i and trading day d , daily turnover is computed as:

$$\text{Turnover}_{i,d} = \frac{\text{Volume}_{i,d}}{\text{SharesOut}_{i,d}},$$

where $\text{Volume}_{i,d}$ is the number of shares traded and $\text{SharesOut}_{i,d}$ is the number of shares outstanding.

Weekly turnover is then constructed by averaging daily turnover over all trading days within week t :

$$\text{Turnover}_{i,t} = \frac{1}{n_t} \sum_{d \in \text{week } t} \text{Turnover}_{i,d},$$

where n_t denotes the number of trading days in week t (typically five).

12- Idiosyncratic Volatility (IVOL).

To measure firm-specific risk on a weekly horizon, we follow the standard factor-residual approach in the idiosyncratic volatility literature (e.g., Ang, Hodrick, Xing, and Zhang, 2006, 2009) while estimating the factor model using a sufficiently long daily window. For each stock i and week t , we estimate the Fama–French three-factor model on daily excess returns over a rolling window of L trading days ($L = 21$) ending at the close of the prior week:

$$r_{i,d}^e = \alpha_{i,t} + \beta'_{i,t} f_d + \varepsilon_{i,d}, d \in \mathcal{W}_{t-1}(L),$$

where $r_{i,d}^e$ is the daily excess return, $f_d = (MKT - RF, SMB, HML)_d$ are the daily factor returns, and $\mathcal{W}_{t-1}(L)$ denotes the set of L trading days used for estimation. Using the estimated coefficients $\hat{\alpha}_{i,t}$ and $\hat{\beta}_{i,t}$, we compute daily residuals during week t ,

$$\hat{\varepsilon}_{i,d} = r_{i,d}^e - \hat{\alpha}_{i,t} - \hat{\beta}'_{i,t} f_d, d \in \text{week } t,$$

and define weekly idiosyncratic volatility as the standard deviation of these within-week residuals:

$$\text{IVOL}_{i,t} = \sqrt{\frac{1}{n_t - 1} \sum_{d \in \text{week } t} (\hat{\varepsilon}_{i,d} - \bar{\hat{\varepsilon}}_{i,t})^2},$$

where n_t is the number of trading days in week t (typically five). This construction avoids estimating the factor model on only a handful of daily observations while producing an idiosyncratic volatility measure that is aligned with the weekly return horizon used in the portfolio and regression tests.

13- Idiosyncratic Skewness (ISKEW).

To capture asymmetry in firm-specific return realizations at the weekly horizon, we construct idiosyncratic skewness using the same factor-residual framework employed for IVOL (e.g., Ang, Hodrick, Xing, and Zhang, 2006, 2009). For each stock i and week t , we rely on the daily residuals from the rolling Fama–French three-factor model estimated over a sufficiently long historical window ending at the close of the prior week. Specifically, using the estimated coefficients $\hat{\alpha}_{i,t}$ and $\hat{\beta}_{i,t}$, we compute daily idiosyncratic return residuals during week t ,

$$\hat{\varepsilon}_{i,d} = r_{i,d}^e - \hat{\alpha}_{i,t} - \hat{\beta}'_{i,t} f_d, d \in \text{week } t.$$

Weekly idiosyncratic skewness is then defined as the standardized third central moment of these within-week residuals:

$$\text{ISKEW}_{i,t} = \frac{1}{n_t} \sum_{d \in \text{week } t} \left(\frac{\hat{\varepsilon}_{i,d} - \bar{\hat{\varepsilon}}_{i,t}}{\text{IVOL}_{i,t}} \right)^3,$$

where n_t denotes the number of trading days in week t and $\bar{\hat{\varepsilon}}_{i,t}$ is the mean residual within the week. By standardizing residuals using weekly idiosyncratic volatility, this measure captures the degree of right-tail asymmetry in firm-specific returns while remaining aligned with the weekly return horizon used throughout the analysis. As with IVOL, this approach avoids estimating higher moments from regressions based on only a handful of daily observations.

14- Return Volatility (Volatility_{21d}).

Short-term return volatility is measured as the realized volatility of daily stock returns over the previous 21 trading days. For each stock i and week t , volatility is computed as:

$$\text{Volatility}_{21d,i,t} = \sqrt{\frac{1}{21-1} \sum_{d=t-21}^{t-1} (R_{i,d} - \bar{R}_{i,t})^2},$$

where $R_{i,d}$ is the daily return of stock i on day d , and $\bar{R}_{i,t}$ is the average daily return over the same 21-day window.

15- Past Returns (PastRet_{21d}).

PastRet_{21d} captures short-horizon momentum or reversal forces and is defined as the cumulative stock return over the previous 21 trading days ending at the close of the prior week. For each stock i and week t ,

$$\text{PastRet}_{21d,i,t} = \sum_{d=t-21}^{t-1} R_{i,d},$$

where $R_{i,d}$ is the daily return of stock i on day d .

Appendix B

Table B1

Summary Statistics for Quintile Portfolios Sorted by Net Retail Buying (Share-Adjusted)

This table reports summary statistics for quintile portfolios formed by sorting stocks on net retail buying adjusted for shares outstanding (NetRetail(SharesOut)). Portfolios are formed weekly from May 2018 to August 2020. In each week, stocks are sorted into five portfolios based on their NetRetail(SharesOut) measured at the end of the previous week, with Portfolio 1 (Low) containing stocks with the lowest net retail buying and Portfolio 5 (High) containing stocks with the highest. For each quintile, the table reports the time-series average of the within-week median values of stock characteristics over the sample period. Reported variables include NetRetail(SharesOut), maximum daily return (MAX), idiosyncratic volatility (IVOL), idiosyncratic skewness (ISKEW), the composite lottery measure *Zscore*, the natural logarithm of market capitalization (SIZE), market beta (BETA), book-to-market ratio (BTM), cumulative return over the 11 weeks prior to portfolio formation excluding the most recent week (MOM), the return in the portfolio formation week (REV), and share turnover (TURN). The final column reports the average number of stocks in each portfolio.

rank	NetRetail(Shares out)	MAX	IVOL	ISKEW	Zscore	BETA	Size	BTM	MOM	REV	TURN	#share
Low	-5.32	2.65	1.38	6.08	-0.45	1.06	5.61	0.92	-1.45	-0.34	15.51	726
2	-0.06	2.90	1.99	9.26	-0.20	0.99	6.71	0.73	-0.80	-0.06	7.36	726
3	0.02	3.25	2.11	6.80	-0.17	0.90	6.96	0.76	0.42	0.19	7.32	726
4	0.16	4.03	3.59	15.57	0.38	1.00	7.01	0.74	2.65	0.50	8.06	726
High	8.91	6.62	3.89	22.28	0.48	1.07	7.16	1.06	9.02	0.67	60.81	726

Table B2

Summary Statistics for Quintile Portfolios Sorted by Abnormal Retail Flow (Size-Price Adjusted)

This table reports summary statistics for quintile portfolios formed by sorting stocks on abnormal retail flow adjusted for firm size and stock price (AbnRetailFlow(P&S)). Portfolios are formed weekly from May 2018 to August 2020. In each week, stocks are sorted into five portfolios based on their AbnRetailFlow(P&S) measured at the end of the previous week, with Portfolio 1 (Low) containing stocks with the lowest abnormal retail flow and Portfolio 5 (High) containing stocks with the highest. For each quintile, the table reports the time-series average of the within-week median values of stock characteristics over the sample period. Reported variables include AbnRetailFlow(P&S), maximum daily return (MAX), idiosyncratic volatility (IVOL), idiosyncratic skewness (ISKEW), the composite lottery measure *Zscore*, the natural logarithm of market capitalization (SIZE), market beta (BETA), book-to-market ratio (BTM), cumulative return over the 11 weeks prior to portfolio formation excluding the most recent week (MOM), the return in the portfolio formation week (REV), and share turnover (TURN). The final column reports the average number of stocks in each portfolio.

rank	AbnRetailFlow(P&S)	MAX	IVOL	ISKEW	Zscore	BETA	Size	BTM	MOM	REV	TURN	#share
Low	-0.31	3.90	3.49	18.05	0.21	0.95	5.80	0.98	5.34	0.66	13.51	726
2	-0.08	3.19	2.27	9.60	-0.10	0.98	6.55	0.76	0.79	0.11	8.41	726
3	-0.01	3.18	2.17	8.87	-0.13	1.05	6.63	0.73	0.38	-0.04	8.66	726
4	0.05	3.30	2.23	8.88	-0.13	1.01	6.93	0.73	0.57	-0.08	9.10	726
High	0.37	5.82	3.28	14.66	0.20	1.00	7.57	1.01	2.81	0.35	59.15	726

Table B3**Summary Statistics for Quintile Portfolios Sorted by Microstructure-Adjusted Abnormal Retail Flow**

This table reports summary statistics for quintile portfolios formed by sorting stocks on microstructure-adjusted abnormal retail flow (AbnRetailFlow(Micro)). Portfolios are formed weekly from May 2018 to August 2020. In each week, stocks are sorted into five portfolios based on their AbnRetailFlow(Micro) measured at the end of the previous week, with Portfolio 1 (Low) containing stocks with the lowest abnormal retail flow and Portfolio 5 (High) containing stocks with the highest. For each quintile, the table reports the time-series average of the within-week median values of stock characteristics over the sample period. Reported variables include AbnRetailFlow(Micro), maximum daily return (MAX), idiosyncratic volatility (IVOL), idiosyncratic skewness (ISKEW), the composite lottery measure *Zscore*, the natural logarithm of market capitalization (SIZE), market beta (BETA), book-to-market ratio (BTM), cumulative return over the 11 weeks prior to portfolio formation excluding the most recent week (MOM), the return in the portfolio formation week (REV), and share turnover (TURN). The final column reports the average number of stocks in each portfolio.

rank	AbnRetailFlow (Micro)	MAX	IVOL	ISKEW	Zscore	BETA	Size	BTM	MOM	REV	TURN	#share
Low	-0.23	4.23	3.57	17.33	0.25	0.97	5.70	0.98	4.37	0.41	14.61	726
2	-0.04	3.19	2.14	8.90	-0.13	0.99	6.57	0.77	-0.02	-0.02	7.83	726
3	-0.01	3.05	1.99	8.79	-0.21	1.00	6.72	0.74	0.41	0.04	7.93	726
4	0.03	3.23	2.12	8.53	-0.14	1.01	7.01	0.73	0.70	0.11	8.82	726
High	0.27	5.76	3.62	16.51	0.28	1.07	7.50	0.99	4.44	0.44	38.08	726

Table B4**Double-Sorted Portfolios on Net Retail Buying (Share-Adjusted) and MAX**

This table reports average future returns for portfolios double-sorted on net retail buying scaled by shares outstanding, NetRetail(SharesOut), and MAX, defined as the maximum daily return during the previous week. Portfolios are formed weekly over the period May 2018 to August 2020. At the beginning of each week, stocks are first sorted into quintiles based on NetRetail(SharesOut), measured at the end of the previous week. Within each net-retail quintile, stocks are then independently sorted into quintiles based on MAX, computed as the highest daily return within the prior week, and held for one week. Panel A reports equal-weighted portfolio returns, while Panel B reports value-weighted portfolio returns. Each cell reports the average weekly excess return, expressed in percent. The rows labeled E[re] report the return spread between the high-MAX and low-MAX portfolios within each NetRetail(SharesOut) quintile. Corresponding Fama–French six-factor alphas (α_{FF6}) and q-factor alphas (α_{QF}) are reported below. Newey–West t-statistics with four lags are shown in parentheses.

	Low MAX	2	3	4	High MAX	E[re]	α_{FF6}	α_{QF}
Panel (A): Equal Weighted								
Low NetRetail(Sharesout)	0.55	0.26	0.18	0.24	0.14	-0.41 (-1.38)	-0.47 (-1.91)	-0.37 (-1.65)
2	0.23	0.22	0.11	-0.08	-0.15	-0.38 (-1.67)	-0.41 (-1.81)	-0.46 (-2.13)
3	0.41	0.16	0.25	0.12	-0.01	-0.42 (-1.28)	-0.42 (-1.57)	-0.48 (-1.77)
4	0.42	0.29	0.30	0.16	0.26	-0.16 (-0.42)	-0.11 (-0.33)	-0.21 (-0.57)
High NetRetail(Sharesout)	0.92	0.75	0.64	0.36	0.02	-0.90 (-1.93)	-0.74 (-1.86)	-0.87 (-1.95)
E[re]	0.37 (1.25)	0.49 (2.42)	0.46 (2.42)	0.12 (0.55)	-0.12 (-0.43)			
α_{FF6}	0.09 (0.46)	0.22 (1.11)	0.24 (1.28)	-0.09 (-0.34)	-0.17 (-0.6)			
α_{QF}	0.38 (1.3)	0.41 (1.81)	0.46 (2.3)	0.18 (0.68)	-0.11 (-0.43)			
Panel (B): Value Weighted								
Low NetRetail(Sharesout)	0.59	0.19	-0.02	0.24	-0.23	-0.82 (-2.01)	-1.10 (-2.67)	-0.76 (-2.19)
2	0.23	0.34	0.05	0.03	0.05	-0.17 (-0.79)	-0.24 (-1.06)	-0.21 (-0.99)
3	0.33	0.15	0.25	0.18	0.31	-0.01 (-0.07)	-0.04 (-0.19)	-0.05 (-0.23)
4	0.32	0.32	0.25	0.15	0.02	-0.30 (-1.01)	-0.35 (-1.19)	-0.45 (-1.39)
High NetRetail(Sharesout)	0.48	0.39	0.29	0.55	0.24	-0.23 (-0.82)	-0.18 (-0.58)	-0.24 (-0.84)
E[re]	-0.12 (-0.59)	0.20 (1.02)	0.31 (1.38)	0.31 (1.12)	0.47 (1.06)			
α_{FF6}	-0.32 (-1.42)	0.12 (0.46)	0.24 (0.88)	0.18 (0.73)	0.60 (1.55)			
α_{QF}	-0.05 (-0.25)	0.12 (0.59)	0.44 (1.68)	0.37 (1.5)	0.48 (1.04)			

Table B5**Double-Sorted Portfolios on Abnormal Retail Flow (Size–Price Adjusted) and MAX**

This table reports average future returns for portfolios double-sorted on abnormal retail trading flow (size–price adjusted), AbnRetailFlow(P&S), and MAX, defined as the maximum daily return during the previous week. Portfolios are formed weekly over the period May 2018 to August 2020. At the beginning of each week, stocks are first sorted into quintiles based on AbnRetailFlow(P&S), measured at the end of the previous week. Within each abnormal retail flow quintile, stocks are then independently sorted into quintiles based on MAX, computed as the highest daily return within the prior week, and held for one week. Panel A reports equal-weighted portfolio returns, while Panel B reports value-weighted portfolio returns. Each cell reports the average weekly excess return, expressed in percent. The rows labeled E[re] report the return spread between the high-MAX and low-MAX portfolios within each AbnRetailFlow(P&S) quintile. Corresponding Fama–French six-factor alphas (α_{FF6}) and q-factor alphas (α_{QF}) are reported below. Newey–West t-statistics with four lags are shown in parentheses.

	Low MAX	2	3	4	High MAX	E[re]	α_{FF6}	α_{QF}
Panel (A): Equal Weighted								
Low AbnRetailFlow(P&S)	0.6	0.25	0.03	0.03	-0.05	-0.65 (-1.99)	-0.66 (-2.54)	-0.67 (-2.52)
2	0.3	0.13	0.05	0.07	-0.06	-0.36 (-1.01)	-0.33 (-1.17)	-0.36 (-1.36)
3	0.38	0.22	0.36	0.11	0.06	-0.31 (-1.03)	-0.29 (-1)	-0.31 (-1.19)
4	0.41	0.29	0.29	0.27	0.15	-0.27 (-1.02)	-0.29 (-1.12)	-0.33 (-1.31)
High AbnRetailFlow(P&S)	0.82	0.67	0.72	0.39	0.25	-0.57 (-1.42)	-0.53 (-1.56)	-0.55 (-1.45)
E[re]	0.21 (0.99)	0.41 (1.86)	0.7 (2.65)	0.36 (1.2)	0.3 (1)			
α_{FF6}	0 (0.01)	0.34 (1.37)	0.4 (1.61)	0.12 (0.47)	0.13 (0.53)			
α_{QF}	0.18 (0.74)	0.36 (1.46)	0.55 (2.01)	0.29 (0.97)	0.29 (0.89)			
Panel (B): Value Weighted								
Low AbnRetailFlow(P&S)	0.38	0.4	0.32	-0.56	-0.01	-0.39 (-1.13)	-0.44 (-1.26)	-0.42 (-1.33)
2	0.64	0.13	0.27	0.04	-0.06	-0.7 (-1.8)	-0.71 (-2.06)	-0.75 (-2.34)
3	0.25	0.11	0.17	0.12	-0.03	-0.28 (-0.91)	-0.35 (-1.21)	-0.34 (-1.14)
4	0.36	0.18	0.51	0.23	0.15	-0.21 (-0.77)	-0.14 (-0.49)	-0.24 (-0.94)
High AbnRetailFlow(P&S)	0.68	0.34	0.14	0.56	0.35	-0.33 (-1.03)	-0.25 (-0.82)	-0.34 (-1.1)
E[re]	0.31 (0.97)	-0.06 (-0.25)	-0.18 (-0.54)	1.13 (2.91)	0.36 (0.93)			
α_{FF6}	0.1 (0.33)	-0.06 (-0.19)	-0.37 (-1.17)	0.8 (2.27)	0.29 (0.86)			
α_{QF}	0.23 (0.84)	-0.11 (-0.47)	-0.2 (-0.6)	1 (2.74)	0.32 (0.81)			

Table B6**Double-Sorted Portfolios on Microstructure-Adjusted Abnormal Retail Flow and MAX****

This table reports average future returns for portfolios double-sorted on microstructure-adjusted abnormal retail trading flow, AbnRetailFlow(Micro), and MAX, defined as the maximum daily return during the previous week. Portfolios are formed weekly over the period May 2018 to August 2020. At the beginning of each week, stocks are first sorted into quintiles based on AbnRetailFlow(Micro), measured at the end of the previous week. Within each abnormal retail flow quintile, stocks are then independently sorted into quintiles based on MAX, computed as the highest daily return within the prior week, and held for one week. Panel A reports equal-weighted portfolio returns, while Panel B reports value-weighted portfolio returns. Each cell reports the average weekly excess return, expressed in percent. The rows labeled E[re] report the return spread between the high-MAX and low-MAX portfolios within each AbnRetailFlow(Micro) quintile. Corresponding Fama–French six-factor alphas (α_{FF6}) and q-factor alphas (α_{QF}) are reported below. Newey–West t-statistics with four lags are shown in parentheses.

	Low MAX	2	3	4	High MAX	E[re]	α_{FF6}	α_{QF}
Panel (A): Equal Weighted								
Low AbnRetailFlow(Micro)	0.54	0.42	0.08	-0.04	0.02	-0.52 (-1.23)	-0.54 (-1.63)	-0.53 (-1.58)
2	0.32	0.15	0.12	0.04	-0.05	-0.37 (-1.02)	-0.28 (-0.94)	-0.41 (-1.24)
3	0.38	0.12	0.25	0.1	0.07	-0.31 (-1.2)	-0.31 (-1.33)	-0.34 (-1.66)
4	0.42	0.23	0.22	0.25	0.13	-0.29 (-1.16)	-0.31 (-1.24)	-0.34 (-1.52)
High AbnRetailFlow(Micro)	0.84	0.44	0.5	0.55	0.27	-0.57 (-1.42)	-0.51 (-1.56)	-0.55 (-1.55)
E[re]	0.3 (2.04)	0.03 (0.13)	0.42 (1.73)	0.6 (2.04)	0.25 (0.92)			
α_{FF6}	0.2 (1.09)	0.06 (0.2)	0.32 (0.93)	0.53 (1.66)	0.23 (0.83)			
α_{QF}	0.21 (1.27)	-0.08 (-0.37)	0.34 (1.25)	0.47 (1.85)	0.19 (0.7)			
Panel (B): Value Weighted								
Low AbnRetailFlow(Micro)	0.42	0.35	0.19	-0.41	-0.18	-0.6 (-1.32)	-0.57 (-1.29)	-0.63 (-1.53)
2	0.37	0.1	0.19	0.25	0.21	-0.16 (-0.57)	-0.27 (-0.87)	-0.33 (-1.13)
3	0.25	0.24	0.26	0.08	0.21	-0.04 (-0.15)	0.03 (0.14)	-0.16 (-0.72)
4	0.32	0.3	0.19	0.32	0.23	-0.08 (-0.4)	-0.07 (-0.28)	-0.21 (-0.8)
High AbnRetailFlow(Micro)	0.71	0.42	0.02	0.64	0.34	-0.36 (-1.22)	-0.36 (-1.17)	-0.34 (-1.19)
E[re]	0.28 (1.42)	0.07 (0.34)	-0.17 (-0.6)	1.05 (3.7)	0.52 (1.15)			
α_{FF6}	0.2 (0.98)	0.16 (0.51)	-0.1 (-0.38)	0.9 (2.63)	0.41 (0.84)			
α_{QF}	0.04 (0.2)	-0.03 (-0.11)	-0.15 (-0.52)	0.91 (3.17)	0.33 (0.75)			

Table B7**Retail Participation and the MAX Effect during Earnings-Announcement and Non-Earnings Weeks****

This table reports double-sorted portfolio tests examining the interaction between retail participation and the MAX anomaly, separately for earnings-announcement (EA) weeks and non-earnings-announcement (NOEA) weeks. Portfolios are formed weekly over the period May 2018 to August 2020. In each week, stocks are first sorted into quintiles based on a retail participation measure measured at the end of the previous week, NetRetail(SharesOut), AbnRetail(P&S), AbnRetailFlow(P&S), or AbnRetailFlow(Micro). Within each retail participation quintile, stocks are then sorted into quintiles based on MAX, defined as the maximum daily return during the previous week, and held for one week. For each retail participation quintile, the table reports the high-minus-low MAX return spread (High MAX – Low MAX). The column labeled E[re] reports the average one-week-ahead excess return spread, while α_{FF6} reports the corresponding alpha from the Fama–French six-factor model. Results are shown for value-weighted portfolios. Panel A reports results for EA weeks, defined as weeks containing a firm's quarterly earnings announcement. Panel B reports results for NOEA weeks. t-statistics, shown in parentheses, are computed using Newey–West standard errors to account for serial correlation. All returns and alphas are expressed in percent per week.

	NetRetail(SharesOut)		AbnRetail(P&S)		AbnRetailFlow(P&S)		AbnRetailFlow(Micro)	
	E[re]	α_{FF6}	E[re]	α_{FF6}	E[re]	α_{FF6}	E[re]	α_{FF6}
Panel (A): Value Weighted Earning Announcement Window								
Low	-0.41	-0.58	-1.40	-1.37	-0.12	-0.21	-0.25	-0.18
	(-0.99)	(-1.19)	(-2.07)	(-2.03)	(-0.49)	(-0.83)	(-0.51)	(-0.40)
2	-0.14	-0.01	-0.60	-0.72	-0.14	-0.21	-0.48	-0.39
	(-0.43)	(-0.02)	(-1.49)	(-2.03)	(-0.55)	(-0.64)	(-0.94)	(-0.87)
3	-0.42	-0.41	-0.02	-0.15	-0.74	-0.72	-0.55	-0.49
	(-1.49)	(-1.51)	(-0.08)	(-0.50)	(-2.26)	(-2.50)	(-1.11)	(-1.09)
4	-0.91	-1.03	-0.85	-0.74	-0.43	-0.40	-0.64	-0.67
	(-3.29)	(-3.11)	(-2.74)	(-3.00)	(-0.75)	(-0.72)	(-1.82)	(-1.84)
High	-0.96	-0.93	-1.06	-1.00	-1.02	-0.96	-0.64	-0.69
	(-1.45)	(-1.73)	(-1.95)	(-2.08)	(-1.66)	(-1.65)	(-0.98)	(-1.26)
Panel (B): Value Weighted NO-Earning Announcement Window								
Low	-0.39	-0.65	-0.65	-0.75	-0.09	-0.04	-0.73	-0.84
	(-1.30)	(-1.89)	(-1.74)	(-1.93)	(-0.55)	(-0.22)	(-1.85)	(-2.05)
2	-0.11	-0.07	0.07	0.09	-0.22	-0.16	-0.11	0.00
	(-0.44)	(-0.27)	(0.28)	(0.30)	(-0.94)	(-0.76)	(-0.30)	(0.00)
3	-0.27	-0.24	0.02	0.13	0.04	-0.09	-0.35	-0.49
	(-1.15)	(-1.07)	(0.10)	(0.48)	(0.17)	(-0.33)	(-1.50)	(-2.09)
4	0.08	-0.03	0.14	0.11	-0.52	-0.66	-0.10	0.02
	(0.36)	(-0.12)	(0.60)	(0.41)	(-1.71)	(-2.09)	(-0.37)	(0.07)
High	-0.13	-0.18	-0.03	-0.09	-0.07	-0.11	0.37	0.39
	(-0.33)	(-0.41)	(-0.09)	(-0.25)	(-0.15)	(-0.25)	(0.83)	(0.88)

Table B8

Value-Weighted Triple-Sorted Portfolios by Size, Abnormal Retail Ownership, and MAX

This table reports value-weighted triple-sorted portfolio results based on firm size, abnormal retail ownership (size-price adjusted), and the MAX characteristic, using weekly data from May 2018 to August 2020. In each week, stocks are first sorted into size terciles (Small, Medium, Big) based on market capitalization measured at the end of the previous week. Within each size tercile, stocks are then sorted into quintiles based on abnormal retail ownership, AbnRetail(P&S), measured at the end of the previous week. Finally, within each size-retail cell, stocks are further sorted into quintiles based on MAX, defined as the maximum daily return during the previous week. Portfolios are held for one week. Panel A, Panel B, and Panel C report results for small, medium, and big firms, respectively. Within each panel, the table reports average weekly excess returns for the resulting portfolios. The rows labeled E[re] report the high-minus-low MAX return spread (High MAX – Low MAX) within each retail ownership category. The rows labeled α_{FF6} and α_{QF} report the corresponding alphas from the Fama–French six-factor model and the Q-factor model, respectively. t-statistics, shown in parentheses, are computed using Newey–West standard errors. All returns and alphas are expressed in percent per week.

PANEL (A): Small Size								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	0.21	0.01	0.07	-0.47	-0.42	-0.63 (-2.15)	-0.62 (-2.16)	-0.58 (-2.09)
2	0.55	0.25	0.28	0.36	-0.03	-0.58 (-1.36)	-0.48 (-1.41)	-0.64 (-1.52)
High AbnRetail(P&S)	0.77	0.24	0.79	0.17	0.03	-0.74 (-1.48)	-0.59 (-1.4)	-0.85 (-1.93)
E[re]	0.55 (1.32)	0.23 (0.48)	0.73 (1.05)	0.64 (1.24)	0.44 (0.81)			
α_{FF6}	0.31 (1.00)	0.31 (0.56)	0.45 (0.74)	0.61 (1.42)	0.35 (0.81)			
α_{QF}	0.58 (1.45)	0.18 (0.41)	0.62 (0.96)	0.64 (1.47)	0.31 (0.7)			
PANEL (B): Medium Size								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	-0.17	0.01	-0.01	-0.16	-0.46	-0.29 (-0.41)	-0.13 (-0.51)	-0.04 (-0.19)
2	0.11	0.16	-0.1	0.11	-0.36	-0.47 (-1.73)	-0.64 (-2.19)	-0.69 (-2.35)
High AbnRetail(P&S)	0.59	0.46	0.34	0.16	0.36	-0.23 (-0.55)	-0.17 (-0.39)	-0.25 (-0.64)
E[re]	0.76 (2.12)	0.45 (1.65)	0.34 (0.96)	0.32 (1.2)	0.82 (2.68)			
α_{FF6}	0.55 (2.67)	0.25 (1.46)	0.19 (0.75)	0.27 (1.33)	0.61 (2.44)			
α_{QF}	0.82 (2.26)	0.45 (1.58)	0.34 (0.99)	0.3 (1.18)	0.60 (2.23)			
PANEL (C): Big Size								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	0.02	-0.01	0.04	0.03	-0.11	-0.13 (-0.41)	-0.07 (-0.39)	-0.1 (-0.66)
2	0.33	0.15	0.15	0.14	0.07	-0.26 (-1.13)	-0.22 (-1.16)	-0.29 (-1.32)
High AbnRetail(P&S)	0.57	0.41	0.26	0.3	0.25	-0.32 (-1.11)	-0.21 (-0.81)	-0.31 (-1.32)
E[re]	0.55 (2.88)	0.42 (2.4)	0.22 (1.76)	0.27 (1.61)	0.36 (1.13)			
α_{FF6}	0.5 (2.3)	0.54 (2.26)	0.2 (1.37)	0.36 (1.97)	0.36 (1.42)			
α_{QF}	0.5 (2.88)	0.43 (2.42)	0.21 (1.77)	0.25 (1.34)	0.29 (1.22)			

Table B9**Value-Weighted MAX Return Spreads Conditional on Firm Size and Retail Participation**

This table reports value-weighted high-minus-low MAX return spreads (High MAX – Low MAX) conditional on firm size and retail participation, following the same triple-sorting procedure as Table 9 and mirroring the structure of Table 10. Using weekly data from May 2018 to August 2020, stocks are first sorted each week into terciles based on market capitalization (Small, Medium, and Big), measured at the end of the previous week. Within each size tercile, stocks are then sorted into quintiles based on retail participation, measured at the end of the previous week using four alternative proxies: NetRetail(SharesOut), AbnRetail(P&S), AbnRetailFlow(P&S), and AbnRetailFlow(Micro). For expositional clarity, the table reports results for three representative retail participation quintiles (Low, 3, and High) within each size group. Within each size–retail cell, stocks are further sorted into quintiles based on MAX, defined as the maximum daily return during the previous week, and held for one week. The table reports the difference in average weekly excess returns between the highest-MAX and lowest-MAX portfolios (E[re]) as well as the corresponding Fama–French six-factor alphas (α_{FF6}). Panels A, B, and C present results for small, medium, and big firms, respectively. t-statistics, shown in parentheses, are computed using Newey–West standard errors. All returns and alphas are expressed in percent per week.

	NetRetail(SharesOut)		AbnRetail(P&S)		AbnRetailFlow(P&S)		AbnRetailFlow(Micro)	
	E[re]	α_{FF6}	E[re]	α_{FF6}	E[re]	α_{FF6}	E[re]	α_{FF6}
PANEL (A): Small Size								
Low	-0.49	-0.41	-0.26	-0.15	-0.54	-0.51	-0.62	-0.6
	(-1.04)	(-0.89)	(-1.11)	(-0.81)	(-1.72)	(-1.99)	(-1.14)	(-1.24)
2	-0.2	-0.23	-0.39	-0.29	-0.09	-0.02	-0.37	-0.38
	(-0.68)	(-0.88)	(-0.93)	(-0.77)	(-0.32)	(-0.11)	(-1.22)	(-1.33)
High	-0.45	-0.32	-0.12	-0.19	-0.47	-0.46	-0.49	-0.33
	(-0.94)	(-0.78)	(-0.34)	(-0.6)	(-1.32)	(-1.45)	(-1.28)	(-1.03)
PANEL (B): Medium Size								
Low	-0.72	-0.61	-0.63	-0.64	-0.42	-0.44	-0.49	-0.48
	(-1.94)	(-2.02)	(-2.72)	(-3.02)	(-1.23)	(-1.62)	(-1.31)	(-1.55)
2	-0.3	-0.28	-0.1	-0.08	-0.27	-0.25	-0.12	-0.12
	(-2.13)	(-2.14)	(-0.34)	(-0.28)	(-1.11)	(-1.03)	(-0.51)	(-0.63)
High	-0.25	-0.33	-0.56	-0.53	-0.26	-0.39	-0.68	-0.69
	(-0.62)	(-0.89)	(-1.29)	(-1.31)	(-1.17)	(-1.68)	(-2.59)	(-2.98)
PANEL (C): Big Size								
Low	-0.61	-0.6	-0.68	-0.71	-0.67	-0.54	-0.55	-0.57
	(-0.96)	(-1.13)	(-2.52)	(-2.69)	(-1.82)	(-2.12)	(-1.18)	(-1.47)
2	-0.57	-0.53	-0.59	-0.61	-0.47	-0.58	-0.4	-0.46
	(-2.15)	(-2.12)	(-1.93)	(-2.22)	(-1.66)	(-1.74)	(-2.39)	(-2.52)
High	-0.83	-0.69	-0.7	-0.45	-0.72	-0.55	-0.29	-0.11
	(-1.82)	(-1.43)	(-1.48)	(-0.94)	(-1.67)	(-1.24)	(-0.77)	(-0.35)

Table B10

Value-Weighted Triple-Sorted Portfolios by Book-to-Market, Abnormal Retail Ownership, and MAX

This table reports value-weighted triple-sorted portfolio results based on book-to-market (BTM), abnormal retail ownership (size–price adjusted), and the MAX characteristic, following the same procedure as Table 11 but using value-weighted returns. Using weekly data from May 2018 to August 2020, stocks are first sorted each week into terciles based on their book-to-market ratio, measured at the end of the previous week, corresponding to Low, Medium, and High BTM firms. Within each BTM tercile, stocks are then sorted into quintiles based on abnormal retail ownership, AbnRetail(P&S), measured at the end of the previous week. Finally, within each BTM–retail cell, stocks are further sorted into quintiles based on MAX, defined as the maximum daily return during the previous week. All portfolios are held for one week. Panel A, Panel B, and Panel C report results for Low, Medium, and High book-to-market firms, respectively. Within each panel, the table reports value-weighted average weekly excess returns for the resulting portfolios. The rows labeled E[re] report the high-minus-low MAX return spread (High MAX – Low MAX) within each retail ownership category. The rows labeled α_{FF6} and α_{QF} report the corresponding alphas from the Fama–French six-factor model and the Q-factor model, respectively. t-statistics, shown in parentheses, are computed using Newey–West standard errors. All returns and alphas are expressed in percent per week.

PANEL (A): Low Book to Market								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	0.08	0.16	0.22	0.19	0.1	0.02 (0.18)	-0.05 (-0.24)	-0.04 (-0.28)
2	0.37	0.16	0.32	0.21	0	-0.36 (-1.09)	-0.39 (-1.19)	-0.42 (-1.33)
High AbnRetail(P&S)	0.71	0.46	0.5	0.08	0.02	-0.69 (-2.05)	-0.52 (-1.61)	-0.69 (-2.15)
E[re]	0.64 (2.6)	0.31 (1.54)	0.28 (1.82)	-0.11 (-0.56)	-0.08 (-0.24)			
α_{FF6}	0.47 (2.3)	0.37 (1.58)	0.21 (1.37)	-0.09 (-0.49)	0 (0.01)			
α_{QF}	0.59 (2.55)	0.34 (1.6)	0.25 (1.72)	-0.15 (-0.72)	-0.06 (-0.19)			
PANEL (B): Medium Book to Market								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	-0.01	-0.18	0.1	-0.01	-0.14	-0.13 (-0.74)	-0.18 (-0.95)	-0.2 (-1.26)
2	0.15	0.06	-0.27	-0.07	0.04	-0.11 (-0.5)	-0.11 (-0.42)	-0.14 (-0.63)
High AbnRetail(P&S)	0.3	0.2	0.15	0.19	0.28	-0.03 (-0.06)	-0.09 (-0.21)	-0.14 (-0.4)
E[re]	0.31 (1.58)	0.38 (2.25)	0.05 (0.42)	0.2 (1.24)	0.42 (1.37)			
α_{FF6}	0.31 (1.42)	0.39 (2.02)	0.02 (0.16)	0.17 (1.12)	0.39 (1.32)			
α_{QF}	0.25 (1.44)	0.3 (1.93)	0.08 (0.55)	0.21 (1.34)	0.31 (1.13)			
PANEL (C): High Book to Market								
	Low-MAX	2	3	4	High-MAX	E[re]	α_{FF6}	α_{QF}
Low AbnRetail(P&S)	0.13	0.04	-0.12	-0.06	-0.29	-0.42 (-1.34)	-0.39 (-1.35)	-0.41 (-1.36)
2	-0.11	0.12	0.09	-0.2	-0.06	0.05 (0.14)	0.07 (0.23)	-0.05 (-0.16)
High AbnRetail(P&S)	0.37	0	-0.16	0.44	-0.25	-0.62 (-1.36)	-0.5 (-1.1)	-0.54 (-1.3)
E[re]	0.24 (0.7)	-0.04 (-0.23)	-0.04 (-0.14)	0.49 (1.45)	0.04 (0.1)			
α_{FF6}	0.28 (0.8)	-0.13 (-0.75)	-0.11 (-0.44)	0.29 (0.85)	0.16 (0.36)			
α_{QF}	0.12 (0.55)	-0.15 (-1.1)	-0.07 (-0.27)	0.37 (1.1)	-0.01 (-0.03)			

Table B11**Value-Weighted MAX Return Spreads Conditional on Book-to-Market and Retail Participation**

This table reports value-weighted high-minus-low MAX return spreads (High MAX – Low MAX) conditional on firm book-to-market ratios and retail participation, following the same empirical design as Table 12 but using value-weighted returns. Using weekly data from May 2018 to August 2020, stocks are first sorted each week into terciles based on their book-to-market ratio (Low, Medium, and High), measured at the end of the previous week. Within each book-to-market tercile, stocks are then sorted into quintiles based on retail participation, measured at the end of the previous week using four alternative proxies: NetRetail(SharesOut), AbnRetail(P&S), AbnRetailFlow(P&S), and AbnRetailFlow(Micro). For expositional clarity, the table reports results for three representative retail participation quintiles (Low, 3, and High) within each book-to-market group. Within each book-to-market–retail cell, stocks are further sorted into quintiles based on MAX, defined as the maximum daily return during the previous week, and held for one week. The table reports the difference in average weekly excess returns between the highest-MAX and lowest-MAX portfolios ($E[re]$) and the corresponding Fama–French six-factor alphas (α_{FF6}). Panels A, B, and C present results for Low, Medium, and High book-to-market stocks, respectively. t-statistics, shown in parentheses, are computed using Newey–West standard errors. All returns and alphas are expressed in percent per week.

	NetRetail(SharesOut)		AbnRetail(P&S)		AbnRetailFlow(P&S)		AbnRetailFlow(Micro)	
	$E[re]$	α_{FF6}	$E[re]$	α_{FF6}	$E[re]$	α_{FF6}	$E[re]$	α_{FF6}
PANEL (A): Low Book to Market								
Low	-0.5	-0.5	0.02	-0.05	-0.21	-0.06	-0.48	-0.46
	(-1.21)	(-1.13)	(0.18)	(-0.24)	(-0.79)	(-0.23)	(-1.17)	(-1.13)
2	-0.29	-0.29	-0.36	-0.39	-0.34	-0.38	-0.43	-0.45
	(-1.04)	(-1.11)	(-1.09)	(-1.19)	(-1.37)	(-1.65)	(-2.09)	(-2.19)
High	-0.68	-0.59	-0.69	-0.52	-0.39	-0.28	-0.55	-0.45
	(-1.97)	(-1.78)	(-2.05)	(-1.61)	(-0.95)	(-0.83)	(-1.45)	(-1.36)
PANEL (B): Medium Book to Market								
Low	-0.21	-0.15	-0.13	-0.18	-0.43	-0.46	-0.37	-0.29
	(-0.41)	(-0.34)	(-0.74)	(-0.95)	(-1.39)	(-1.43)	(-0.95)	(-0.82)
2	-0.07	-0.12	-0.11	-0.11	-0.34	-0.24	-0.07	-0.10
	(-0.39)	(-0.54)	(-0.5)	(-0.42)	(-1.17)	(-1.1)	(-0.32)	(-0.39)
High	0.31	-0.01	-0.03	-0.09	0.59	0.45	0.22	0.13
	(0.8)	(-0.01)	(-0.06)	(-0.21)	(1.88)	(1.29)	(0.71)	(0.45)
PANEL (C): High Book to Market								
Low	-0.11	-0.04	-0.42	-0.39	0.05	0.22	-0.38	-0.28
	(-0.24)	(-0.08)	(-1.34)	(-1.35)	(0.09)	(0.46)	(-0.77)	(-0.6)
2	-0.07	-0.19	0.05	0.07	-0.69	-0.77	-0.32	-0.54
	(-0.29)	(-0.7)	(0.14)	(0.23)	(-2.38)	(-2.53)	(-0.95)	(-1.41)
High	-0.6	-0.36	-0.62	-0.50	-0.01	-0.15	-0.48	-0.37
	(-0.96)	(-0.68)	(-1.36)	(-1.1)	(-0.02)	(-0.44)	(-0.98)	(-0.86)